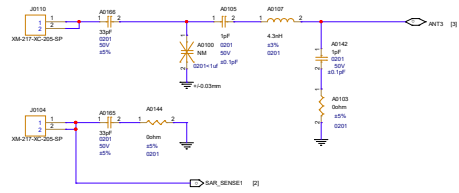
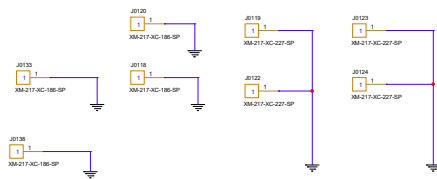


ANT3_MHB



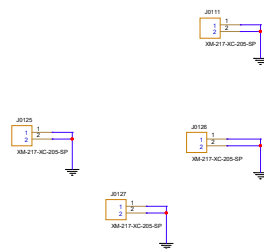
螺丝垫片



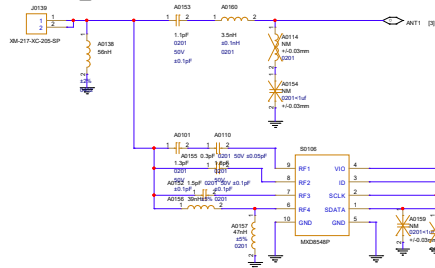
线卡



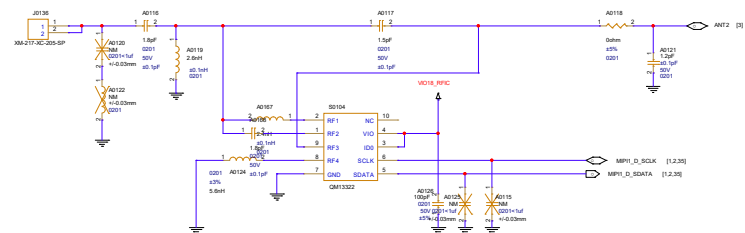
主板接地



ANT1_LB



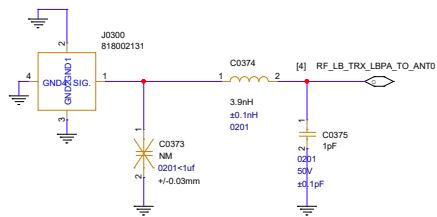
ANT2_MHB



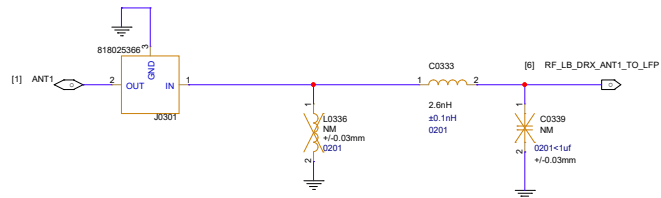
DECO 接地



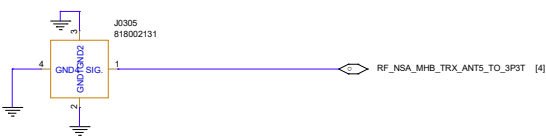
ANT0
LB_TRX
B32/N75_PRX



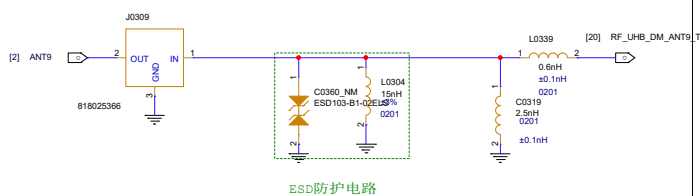
ANT1
LB_DRX
N78_DM



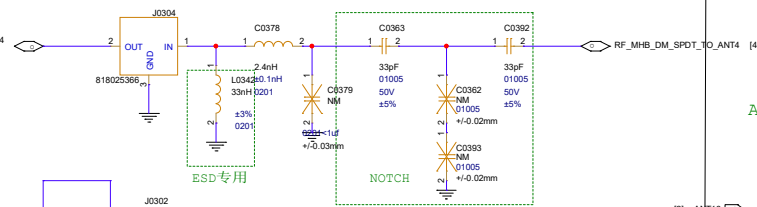
ANT5 MHB_PM



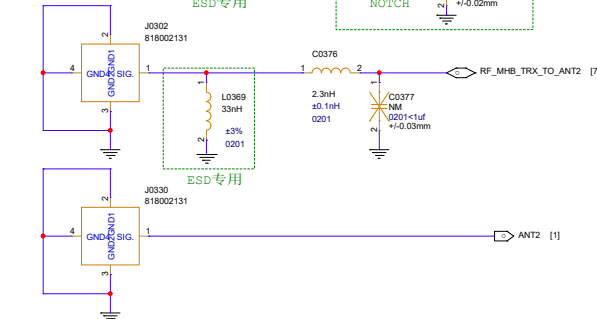
ANT9 N77/78_DM



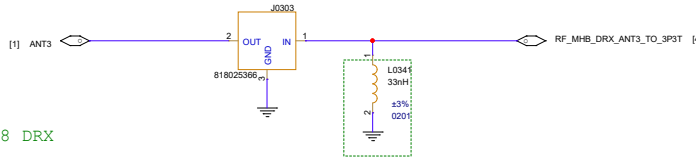
ANT4 MHB_DM



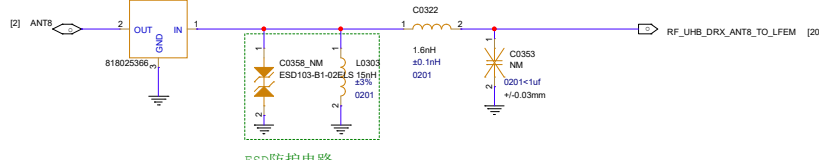
ANT2 MHB_TRX



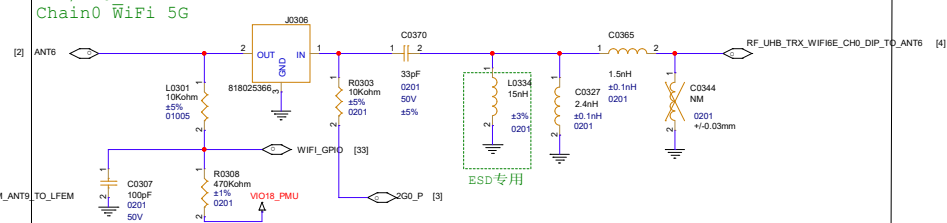
ANT3 MHB_DRX

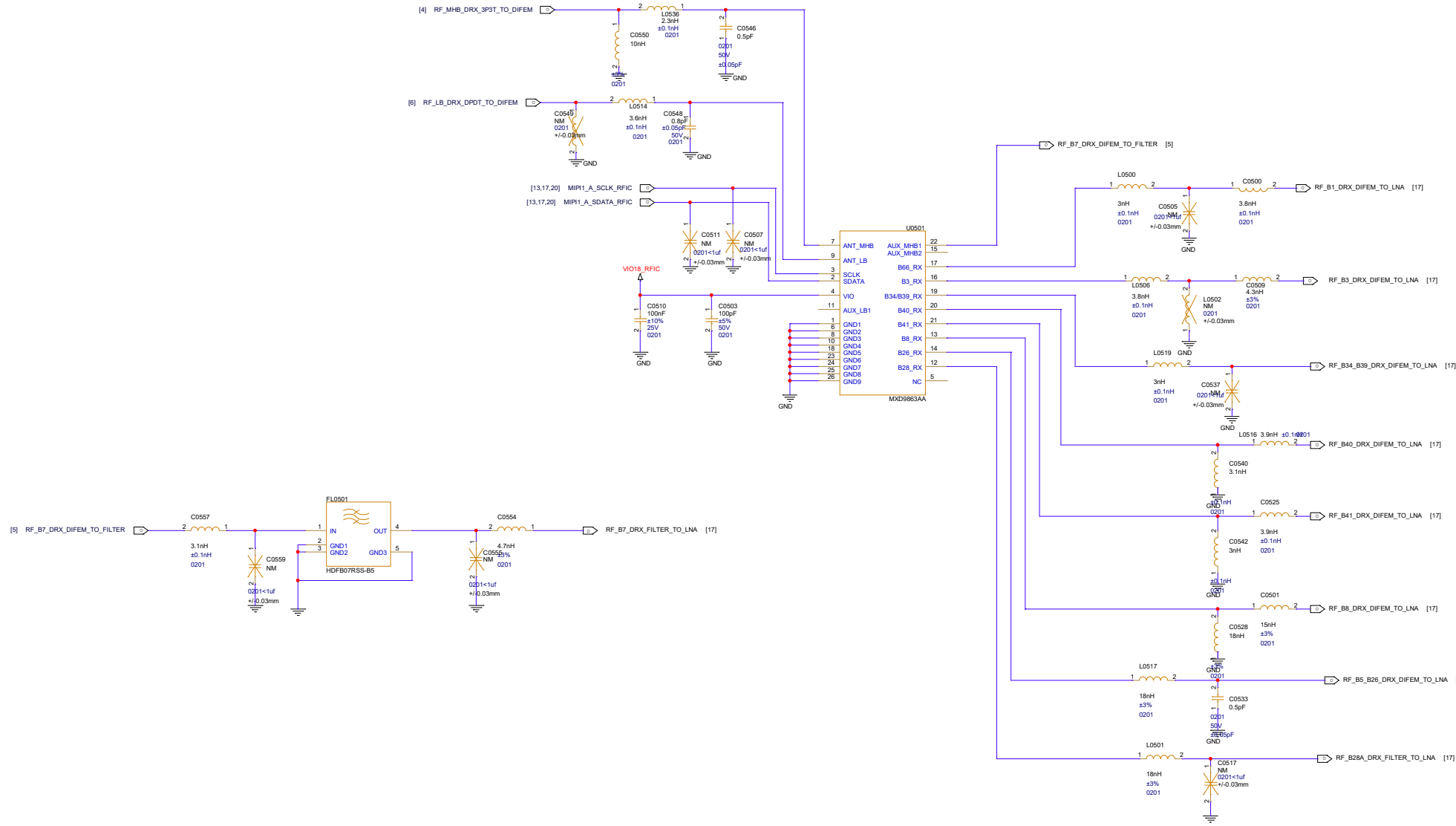


ANT8
N77/78 DRX

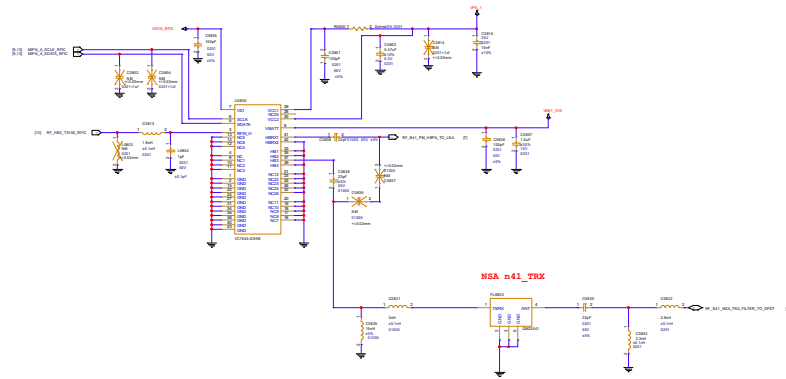


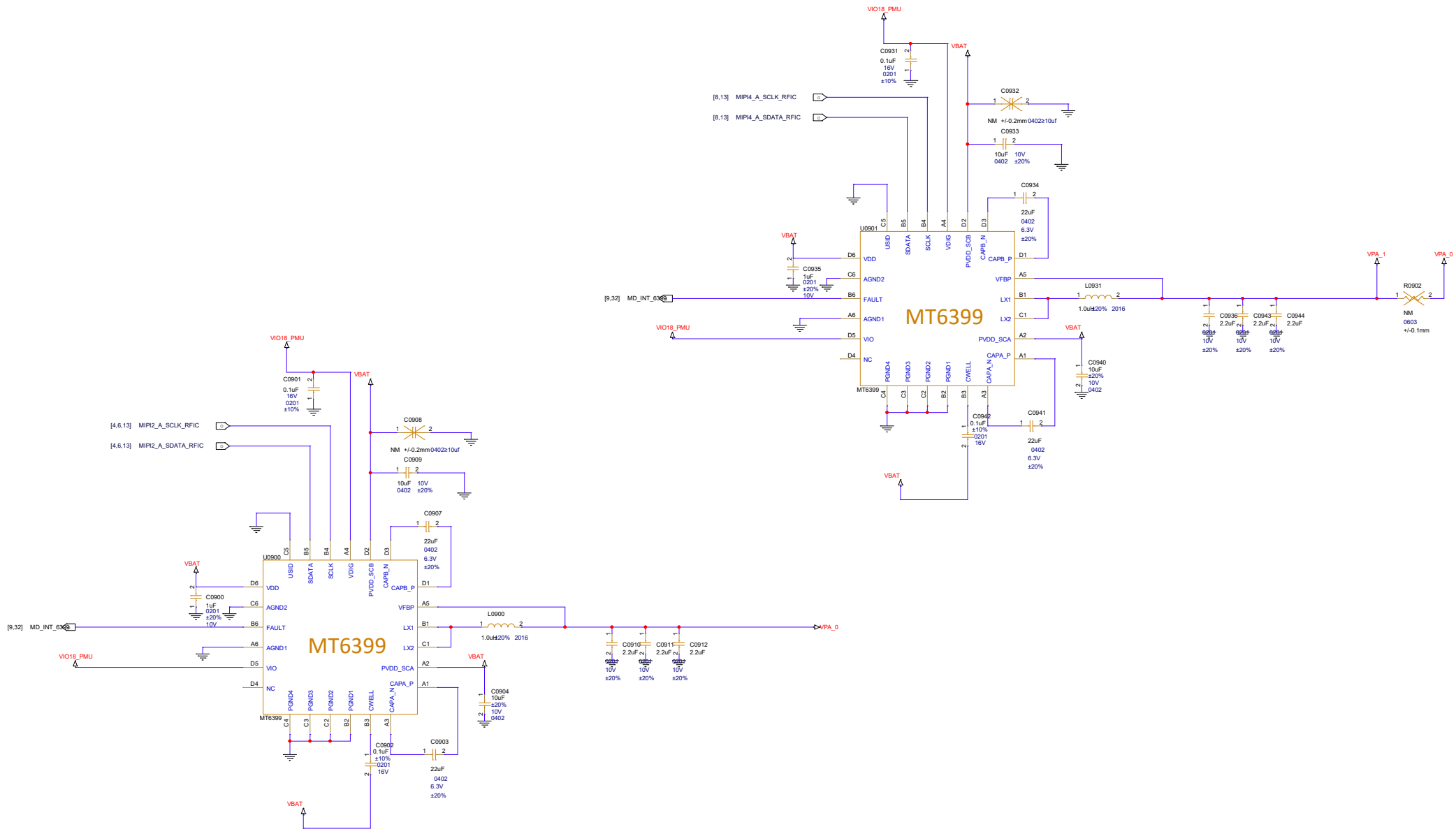
ANT6
N77/N78_TRX
Chain0 WiFi 5G

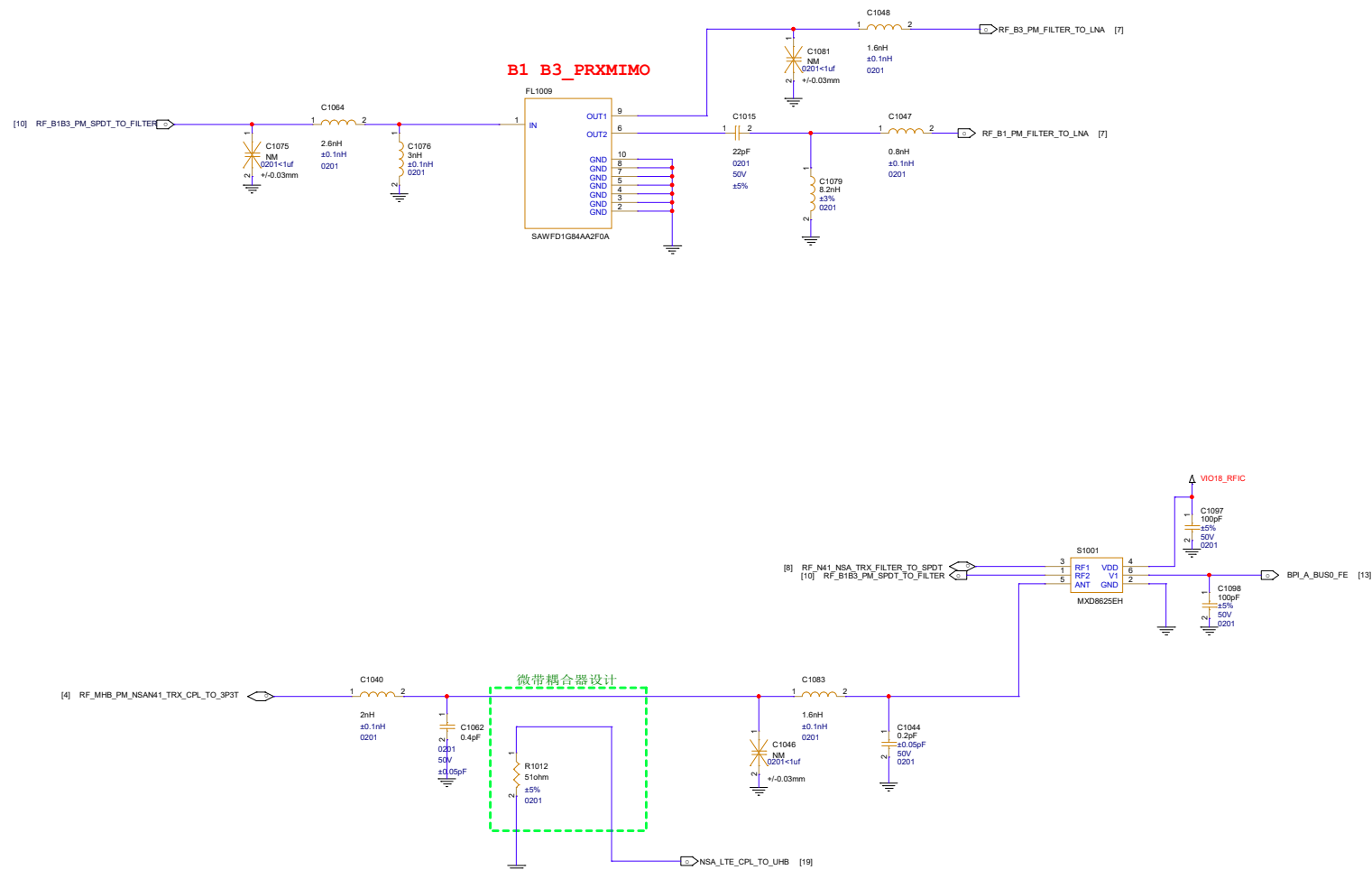




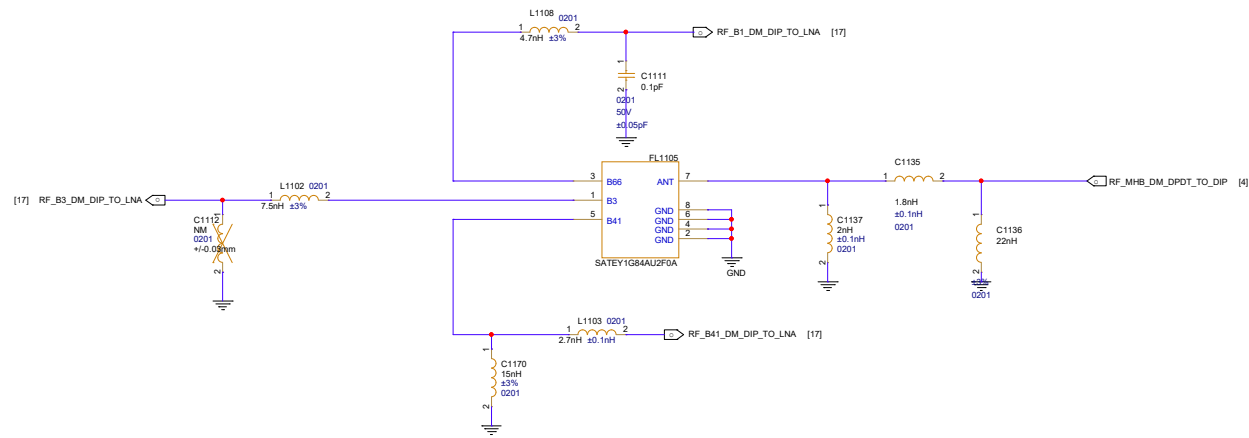
ENDC PA

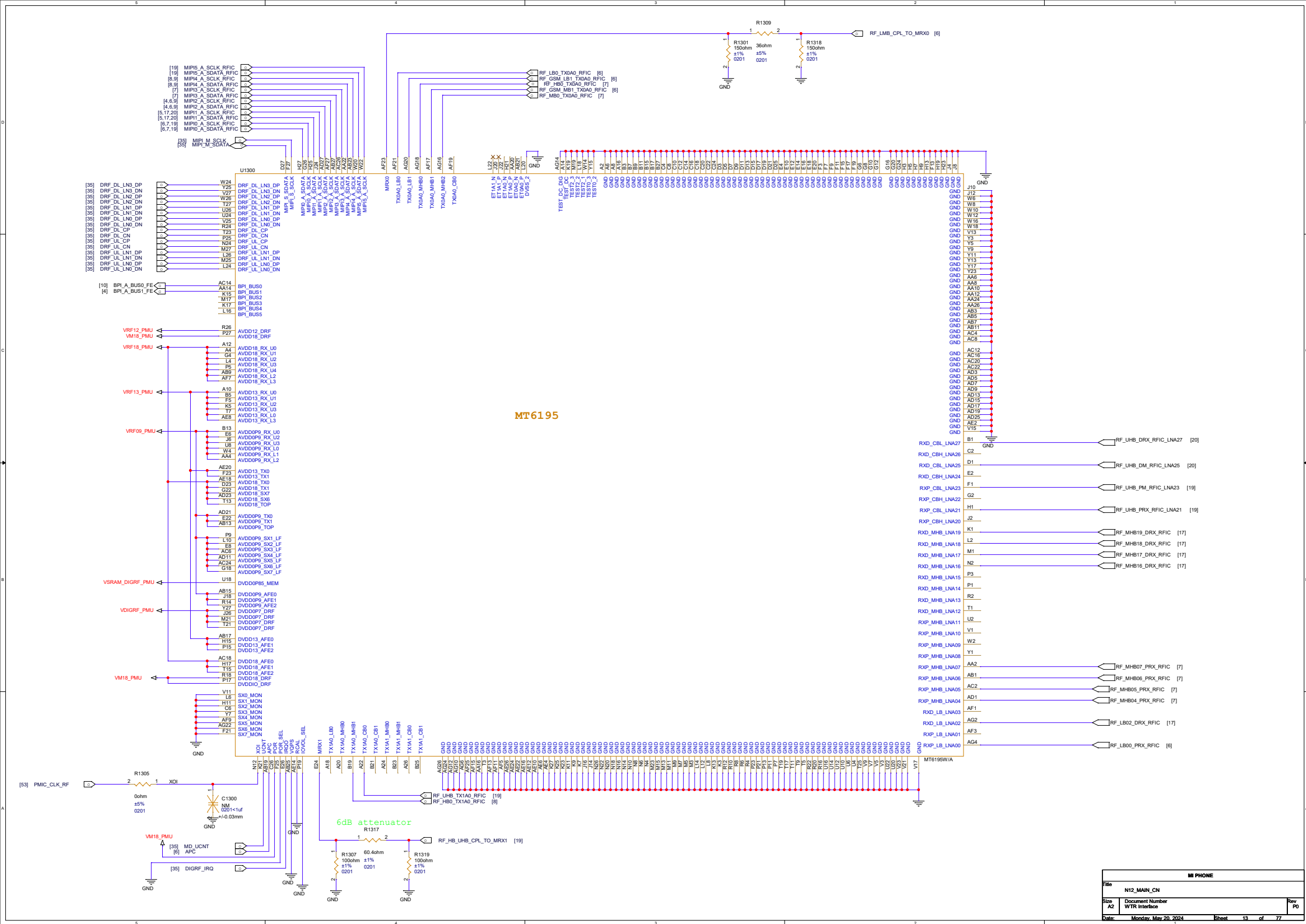




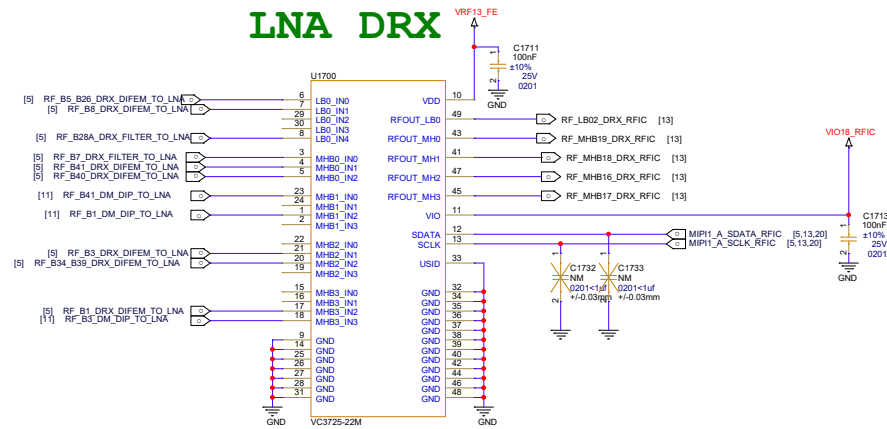


B1_B3_B41_DRX MIMO

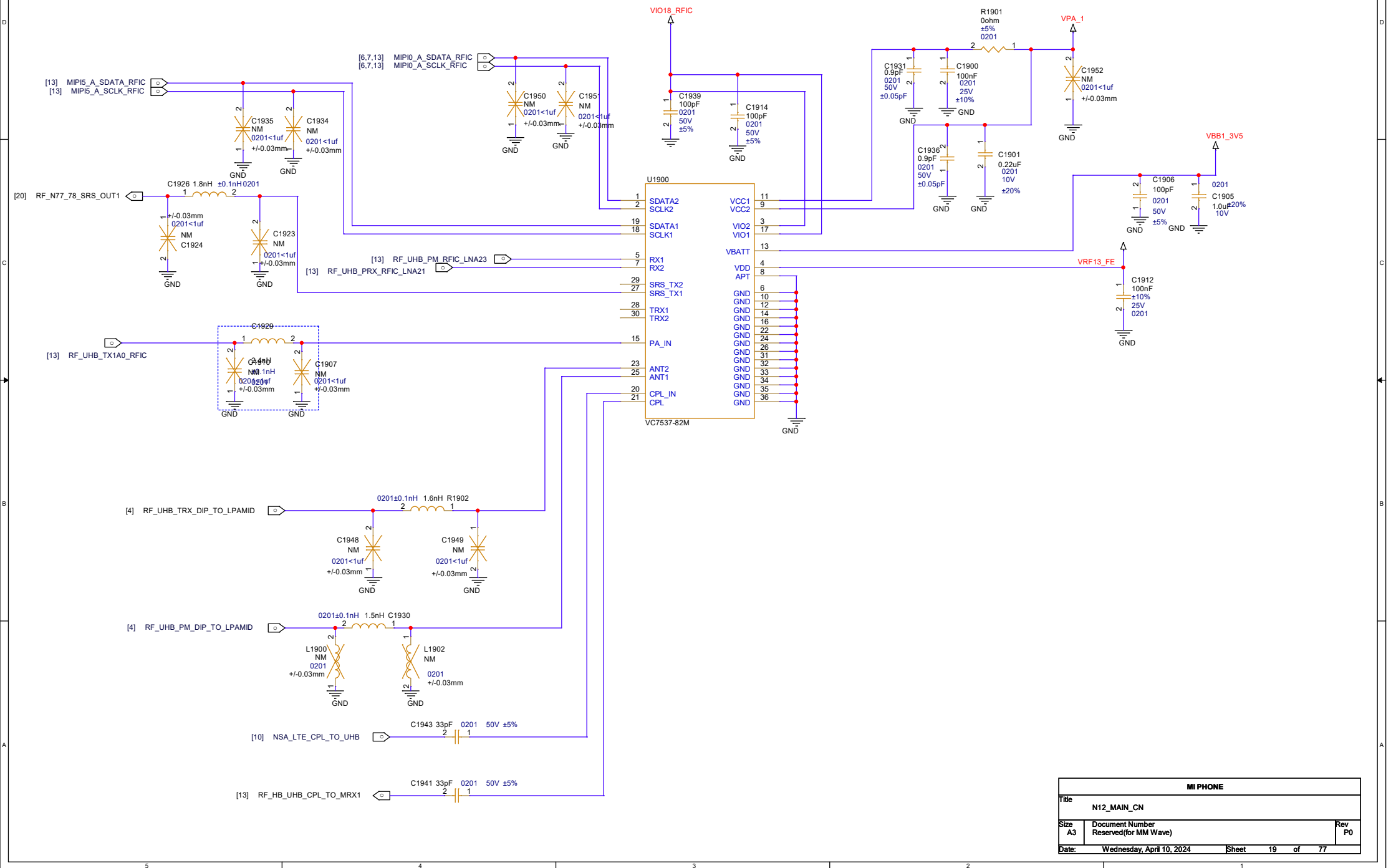


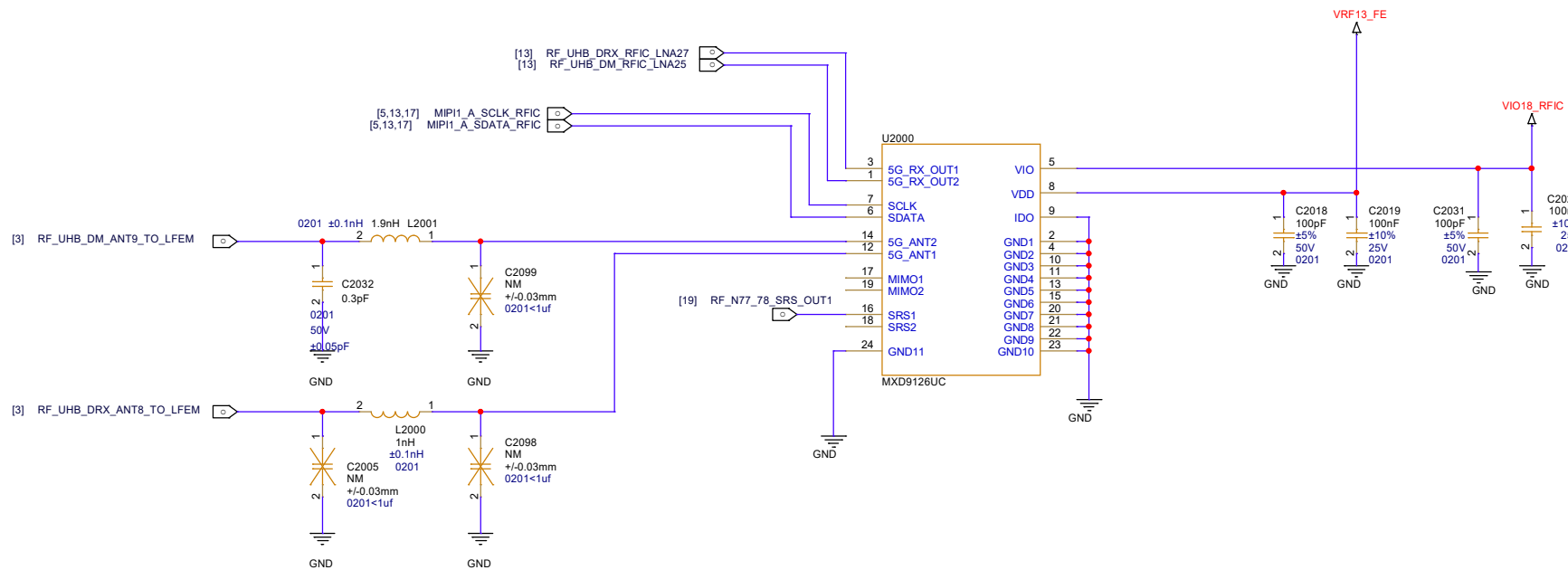


LNA DRX

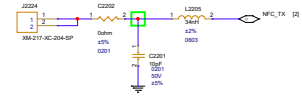
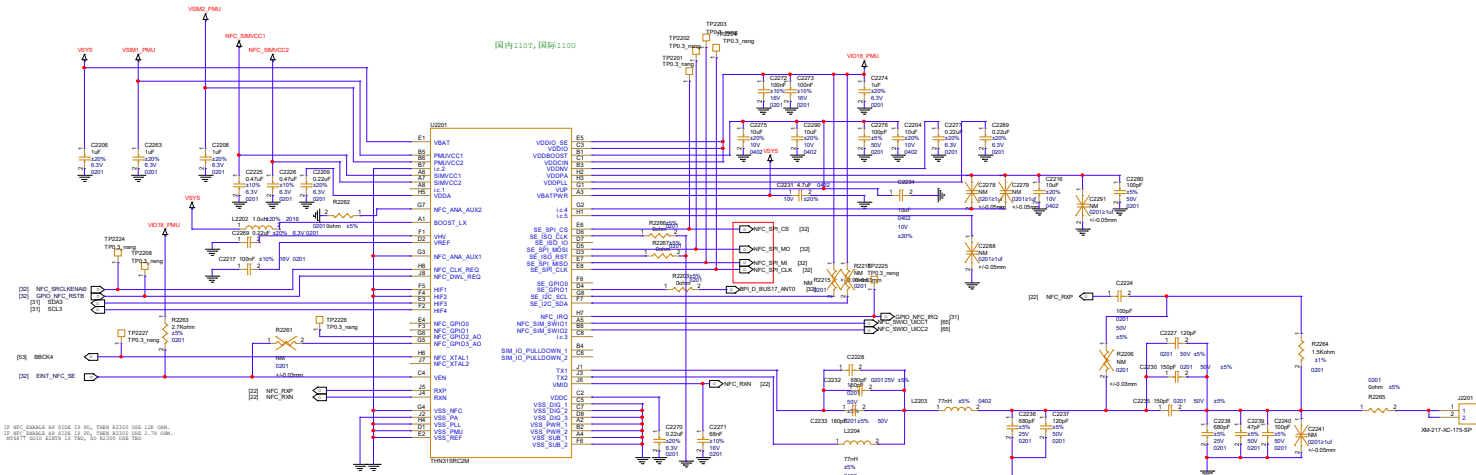


NR N77/N78
LPAMID

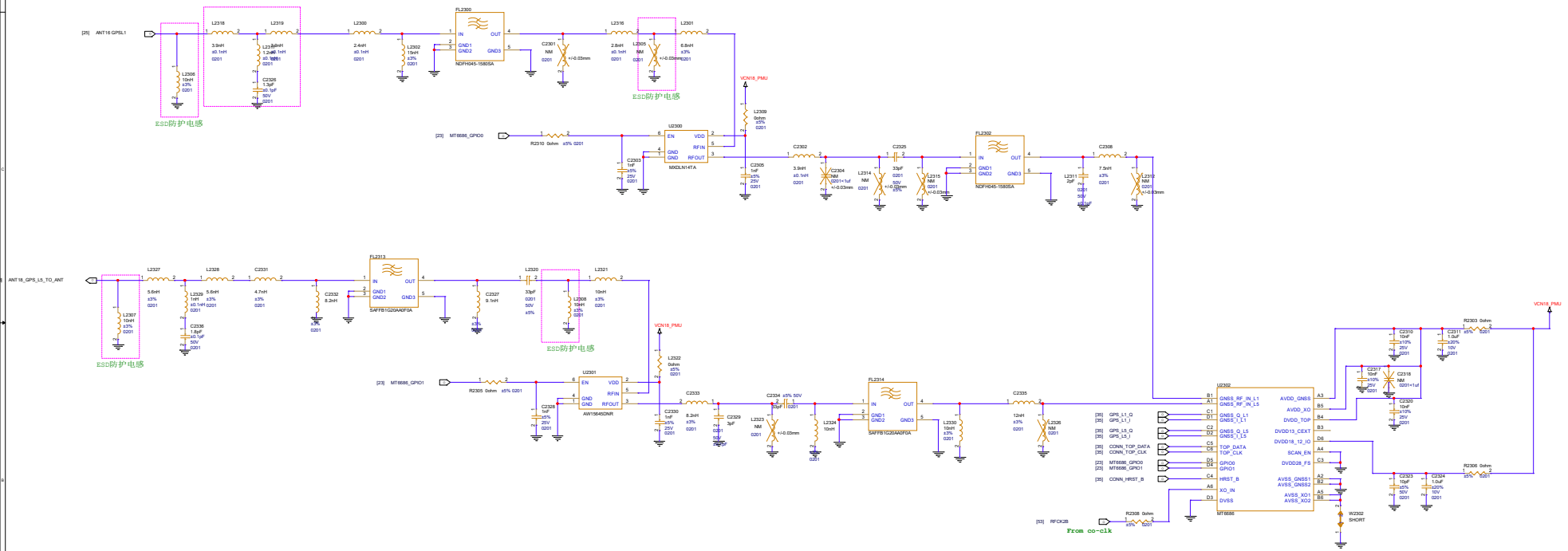


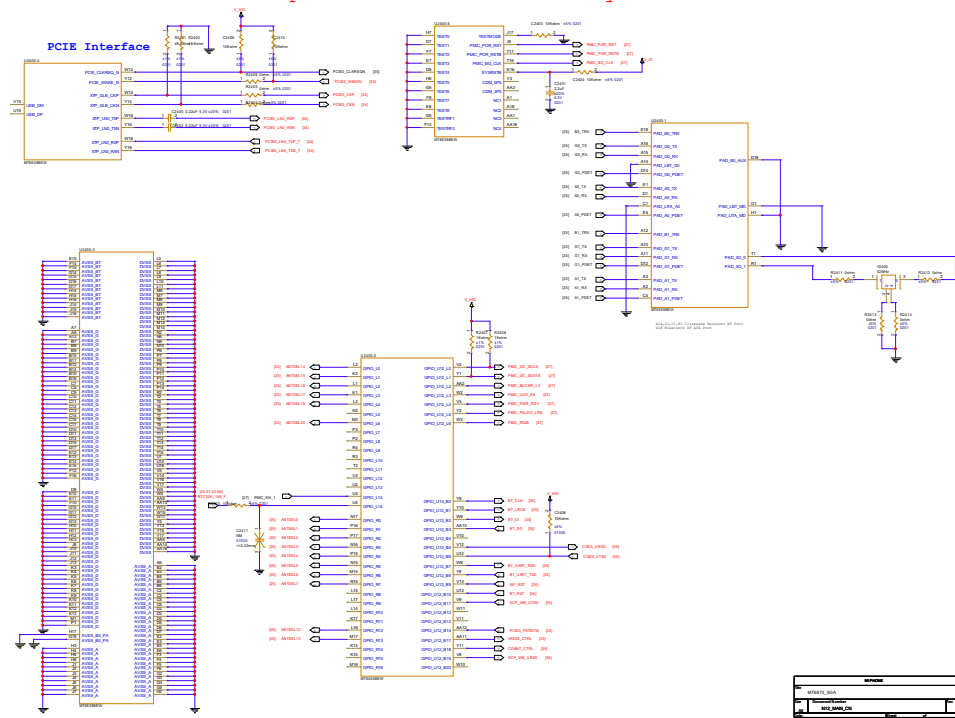


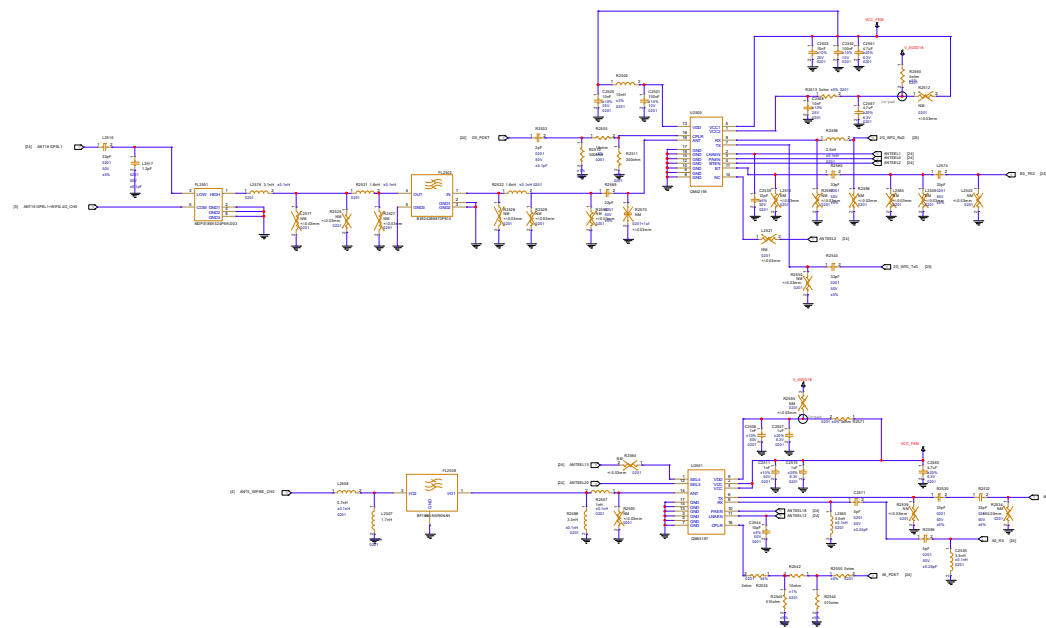
MI PHONE		
Title		
N12_MAIN_CN		
Size	Document Number	Rev
A3	Reserved (for 5G-WTR I)	P0
Date:	Wednesday, April 10, 2024	Sheet 20 of 77



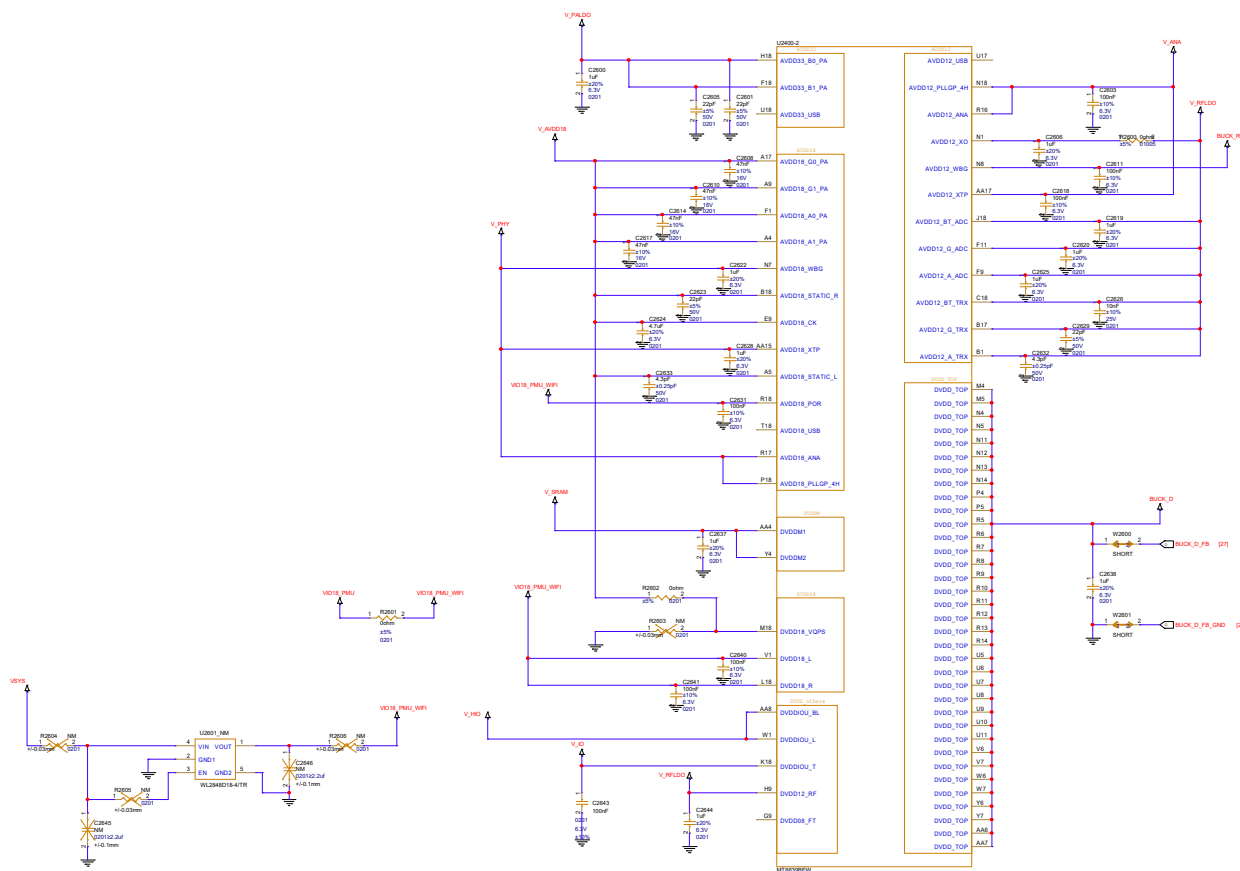
Pre-released for BOM and PCB placement evaluation only

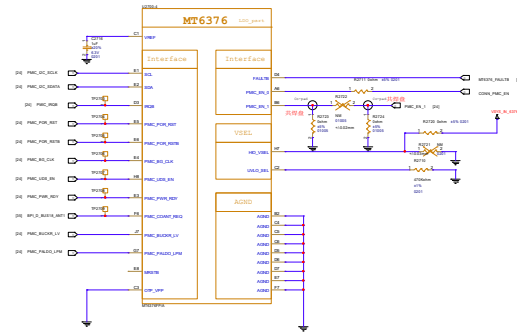
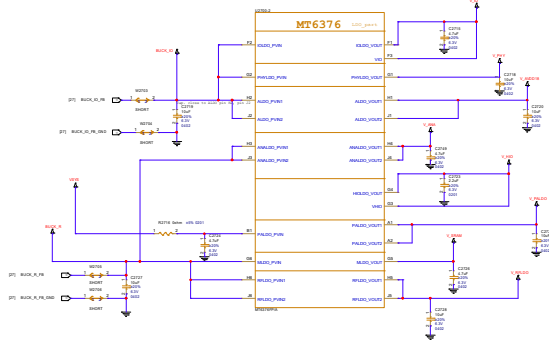
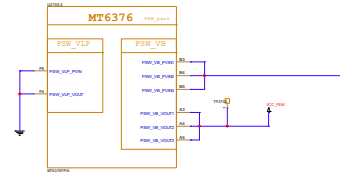
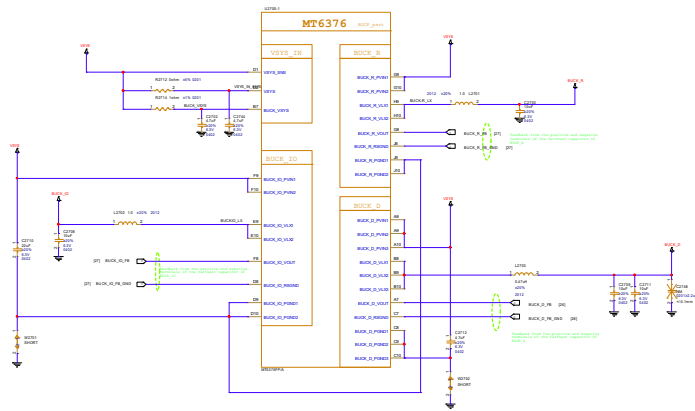




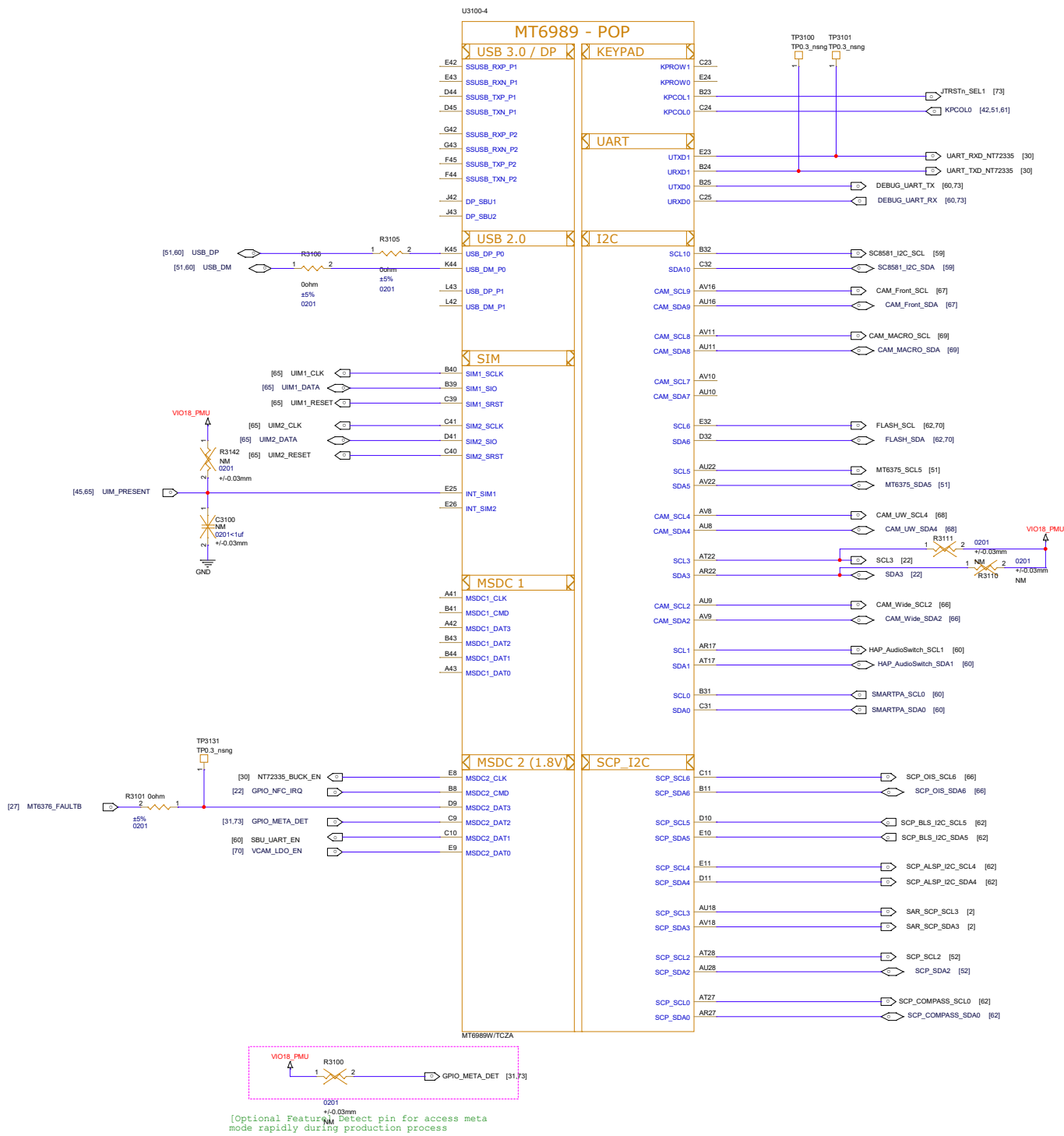


Pre-released for BOM and PCB placement evaluation only

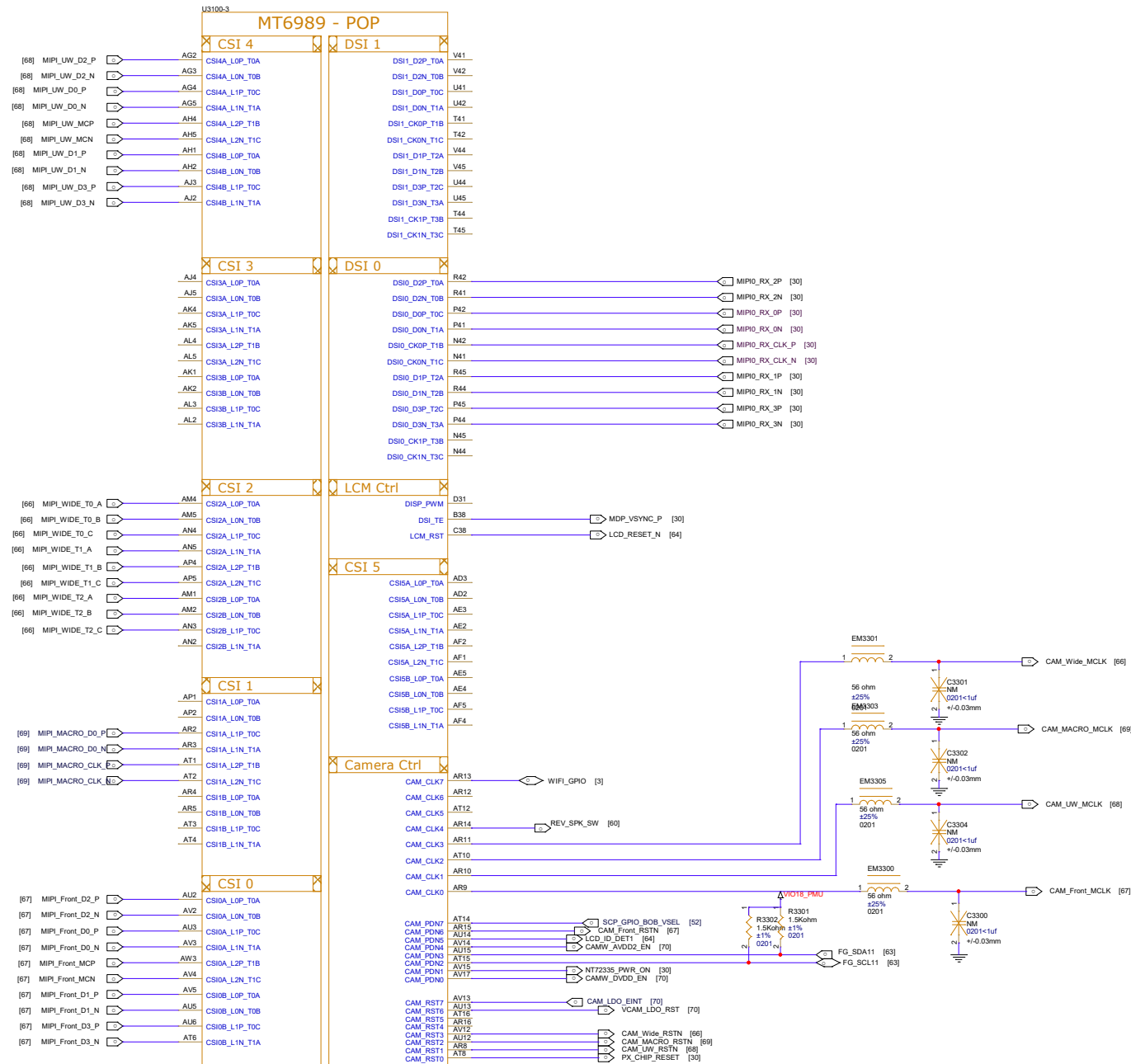


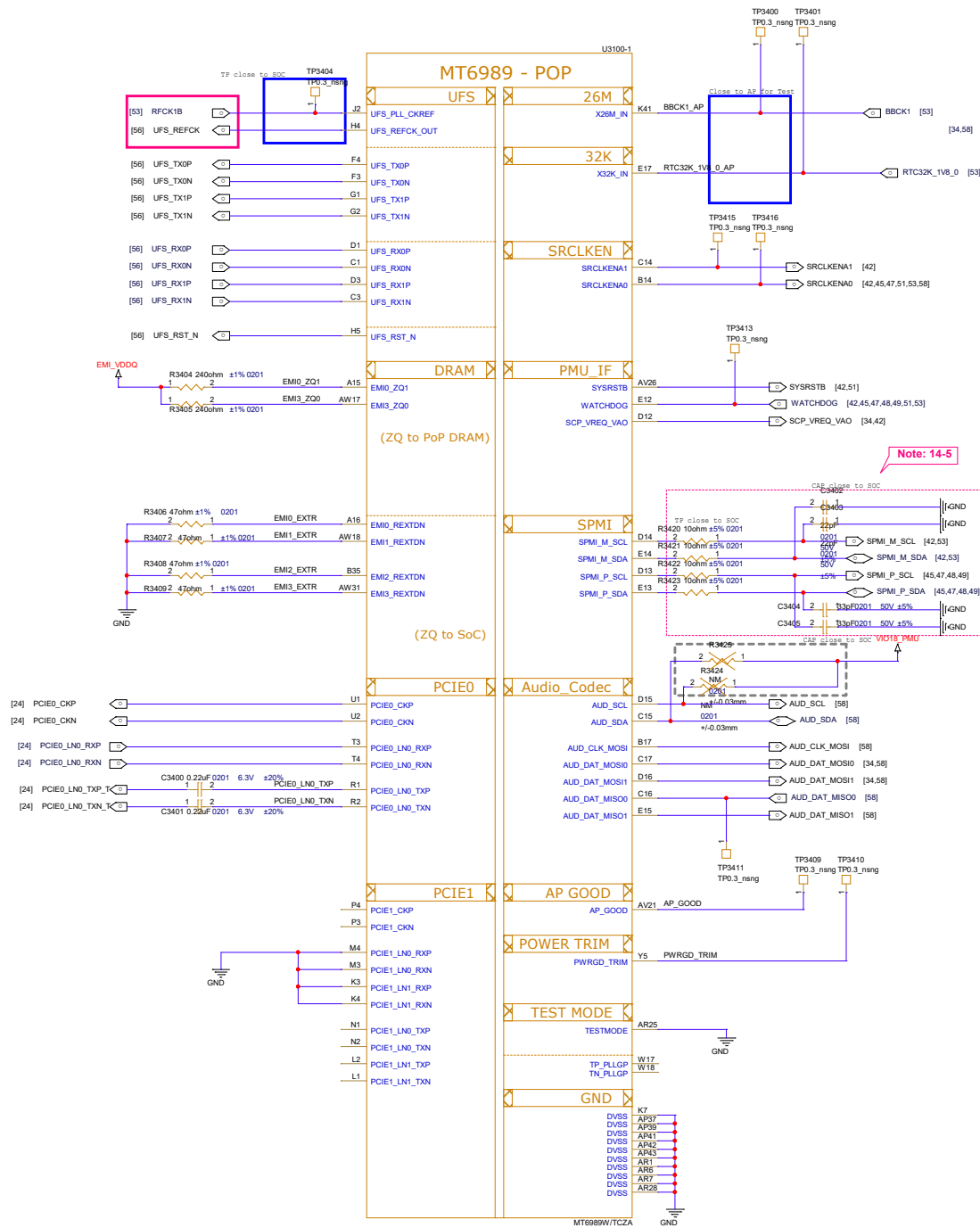


Revision			
Rev	001	001	001
Rev	001	001	001
Rev	001	001	001

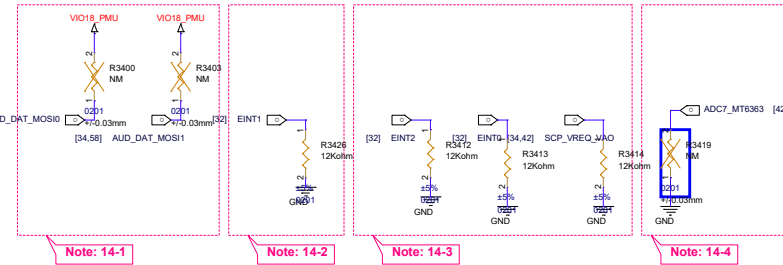


BLS
ALSP
COMPASS





[Optional Feature] Detect pin for access meta mode rapidly during production process



Schematic design notice:

Note 14-1: "AUD_DAT_MOSI0" and "AUD_DAT_MOSI1" pin features in pin strapping to enable JTAG.

AUD_DAT_MOSI0	AUD_DAT_MOSI1	AP JTAG	IO JTAG
L (Default)	L (Default)	N/A	N/A
L	H (by external PU)	SPI0_CLK, SPI0_CSB SPI0_MO, SPI0_MI KPCOL1	N/A
H (by external PU)	L	SPI0_CLK, SPI0_CSB SPI0_MO, SPI0_MI KPCOL1	TDM_MCLK, TDM_BCK TDM_LRCK, TDM_DATA0 TDM_DATA1
H (by external PU)	H (by external PU)	MSDC1_CLK, MSDC1_CMD MSDC1_DAT0, MSDC1_DAT1 MSDC1_DAT2	N/A

Note 14-2: "EINT1" ball is pin strapping to select DRAM Type for LP5X device.

EINT1 (GPIO1)	DRAM Type
L (Default)	LP5X (8533Mbps)
H (by external PU)	LP5X (7500Mbps)

Note 14-3: Pin straps

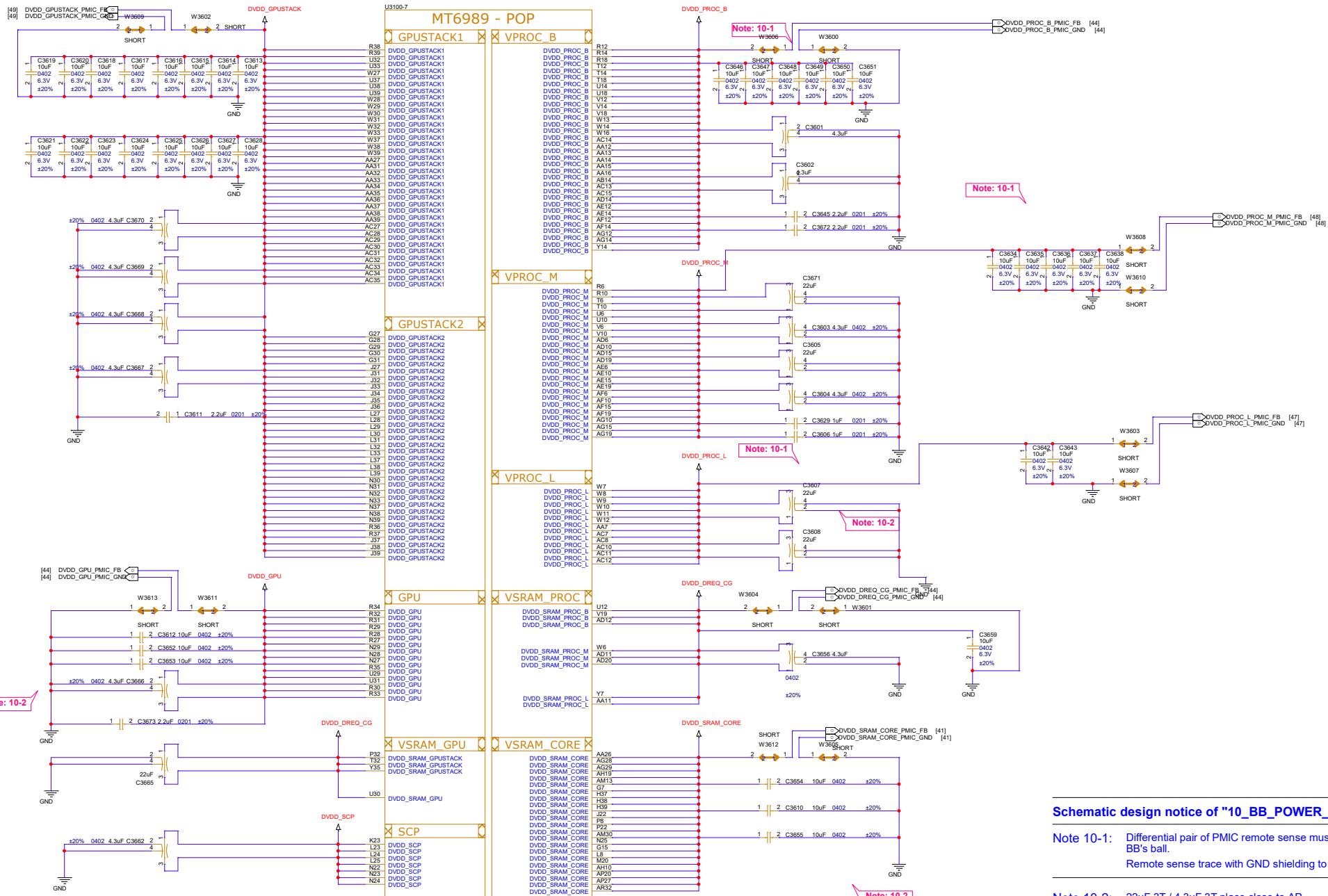
EINT2 (GPIO2)	EINT0 (GPIO0)	SCP_VREQ_VAO (GPIO167)	Mode
L (Default)	L (Default)	L (Default)	UFS4.0-LS
L (Default)	L (Default)	H (by external PU)	UFS4.0-HS
L (Default)	H (by external PU)	L (Default)	UFS3.1 w/ 512M
L (Default)	H (by external PU)	H (by external PU)	UFS3.1 w/ 26M
H (by external PU)	L (Default)	L (Default)	Reserved
H (by external PU)	L (Default)	H (by external PU)	Reserved

Note 14-4: "ADC7_MT6363" pin features in pin strapping to identify Samsung LP5x.

ADC7_MT6363	DRAM Vendor	Note
L (By ext 0R PD)	SS	For SS >= 7500
Floating	non-SS	For other vendors

Note 14-5: "SPMI_M_SCL, SPMI_M_SDA, SPMI_P_SCL, SPMI_P_SDA" add R1437~R1440 and reserved C1403~C1406 for signal quality. Please refer to MTK check list.

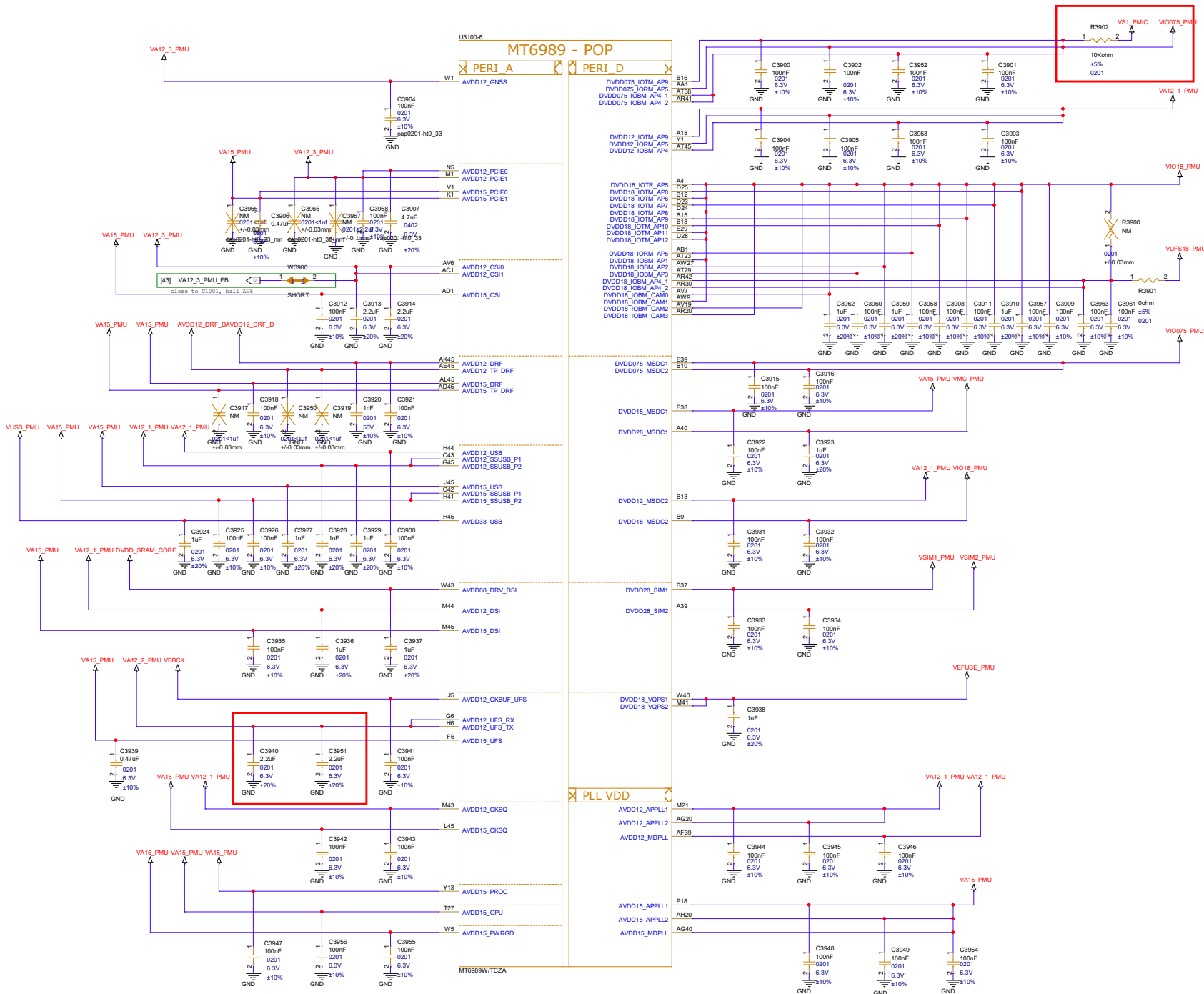
Note: 10-2



Schematic design notice of "10_BB_POWER_PDN" page.

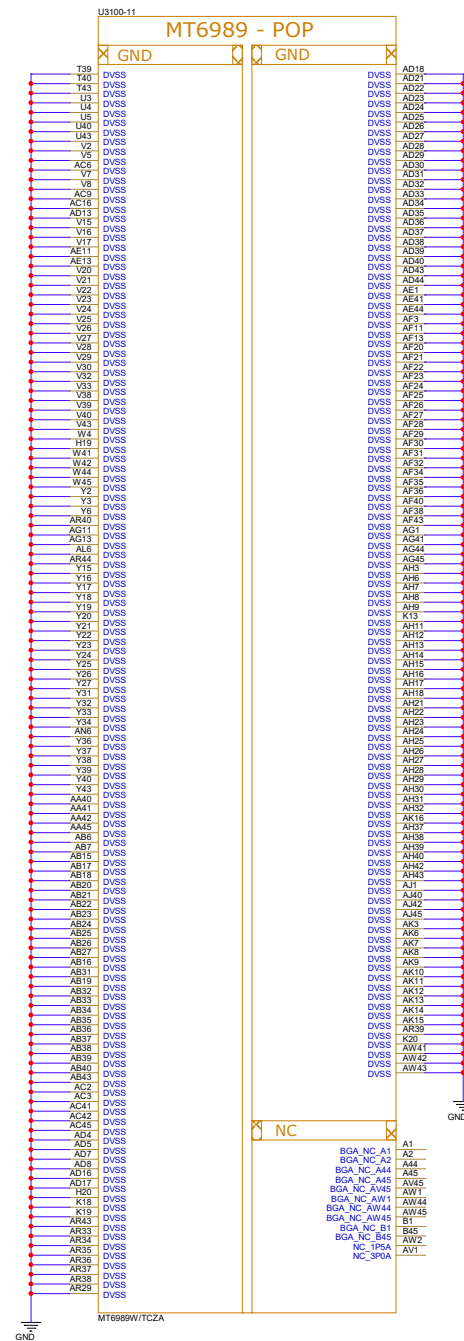
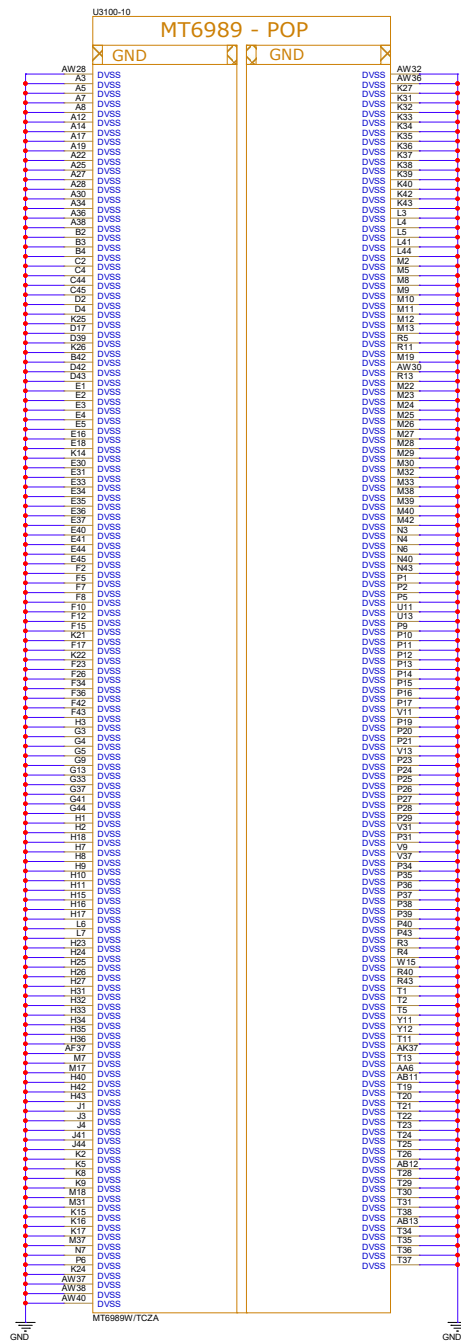
Note 10-1: Differential pair of PMIC remote sense must be close to BB's ball.
Remote sense trace with GND shielding to PMIC (Differential)

Note 10-2: 22uF 3T / 4.3uF 3T place close to AP

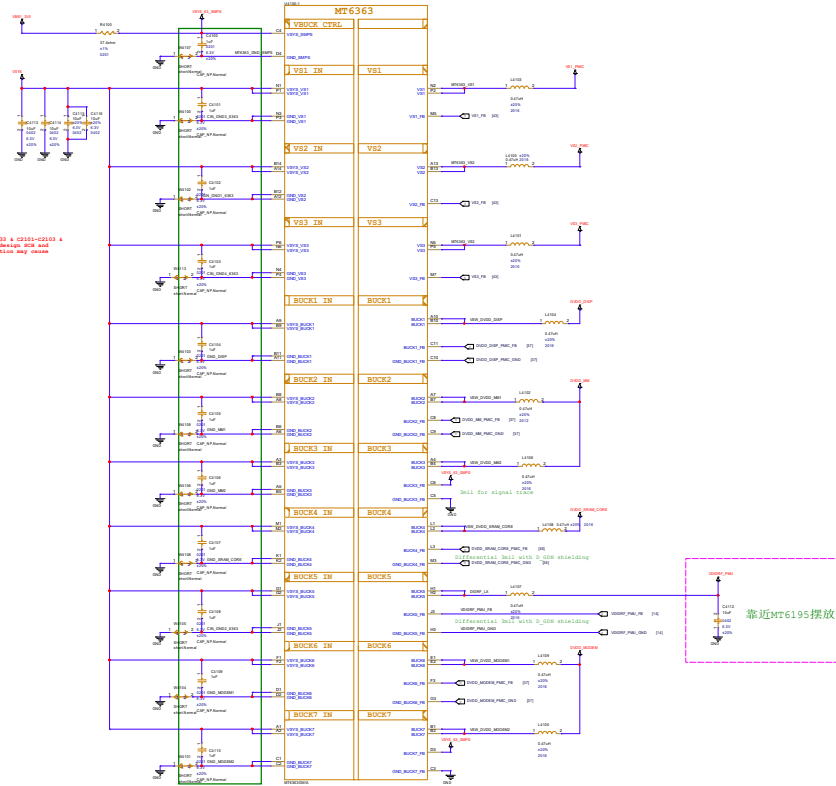


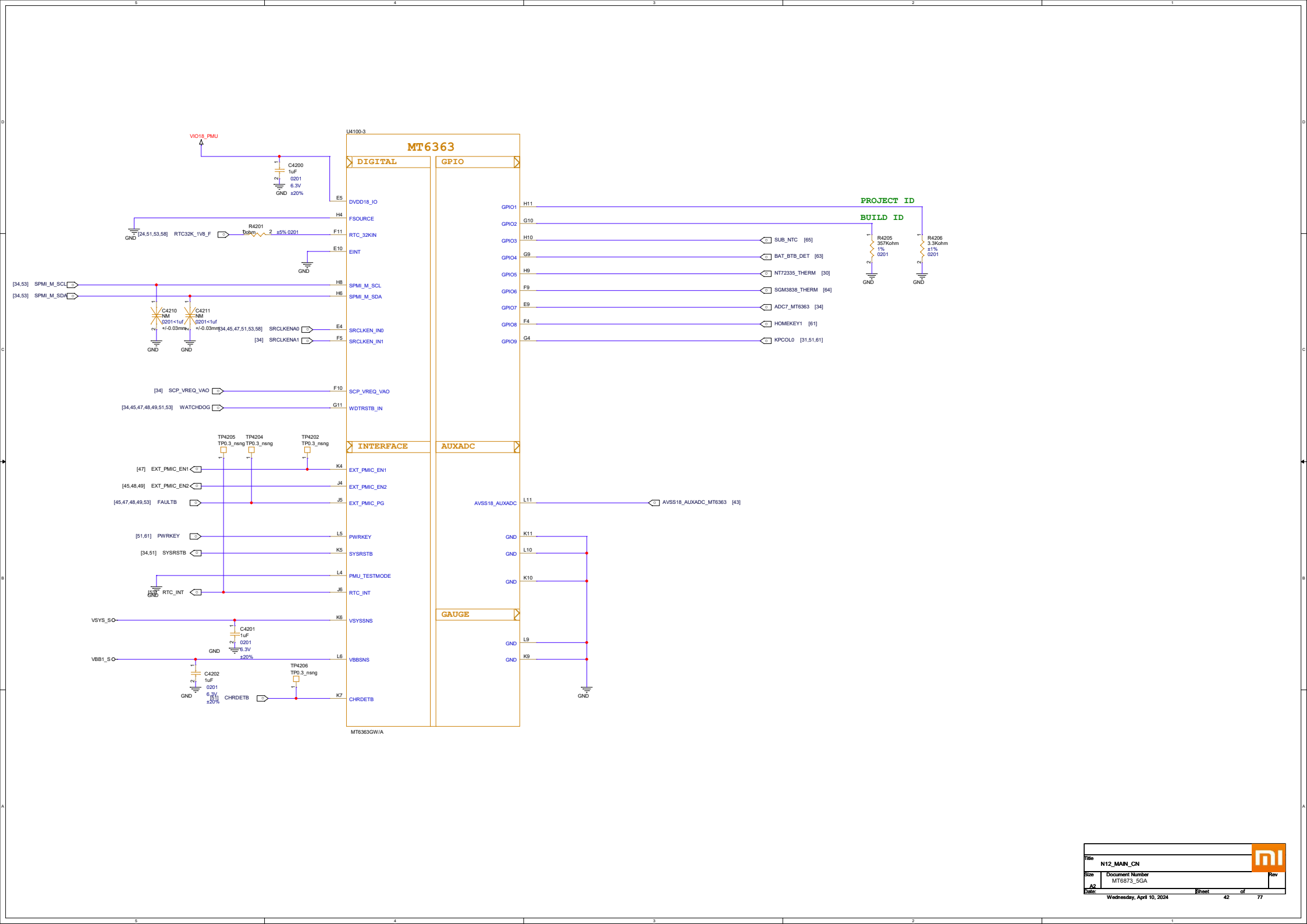
Schematic design notice:
Note 13-1:
When PCIe0 function not used, AVDDxx_PCIEx stabilizing capacitor(C1309/C1311/C1312) can be NC or removed.
When PCIe1 function not used, AVDDxx_PCIEx stabilizing capacitor(C1365/C1308/C1310) can be NC or removed.

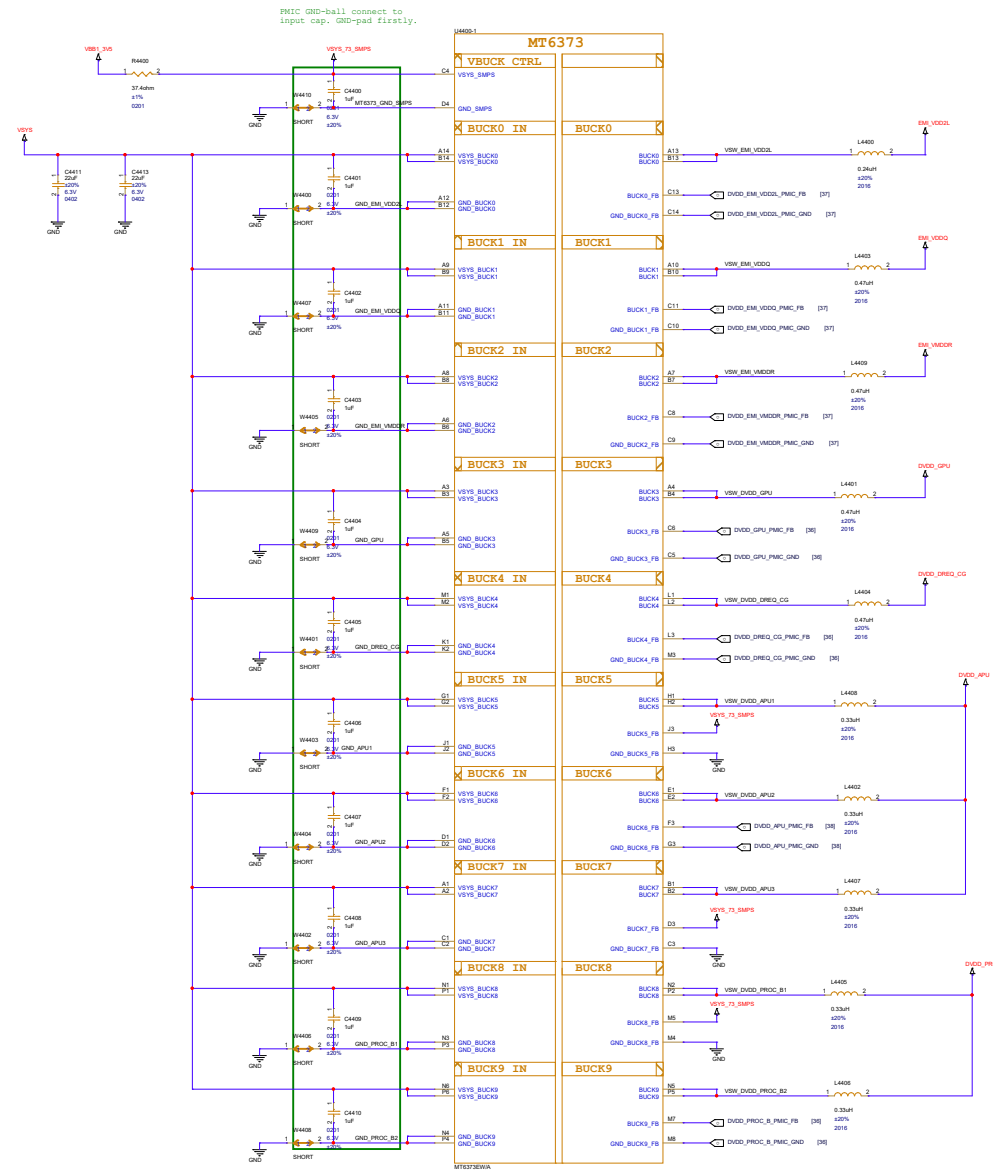
Title		MT6873_5GA	
Size	Document Number		
A2	N12_MAIN_CN		
Date	Wednesday, April 10, 2024	Sheet 39 of 77	



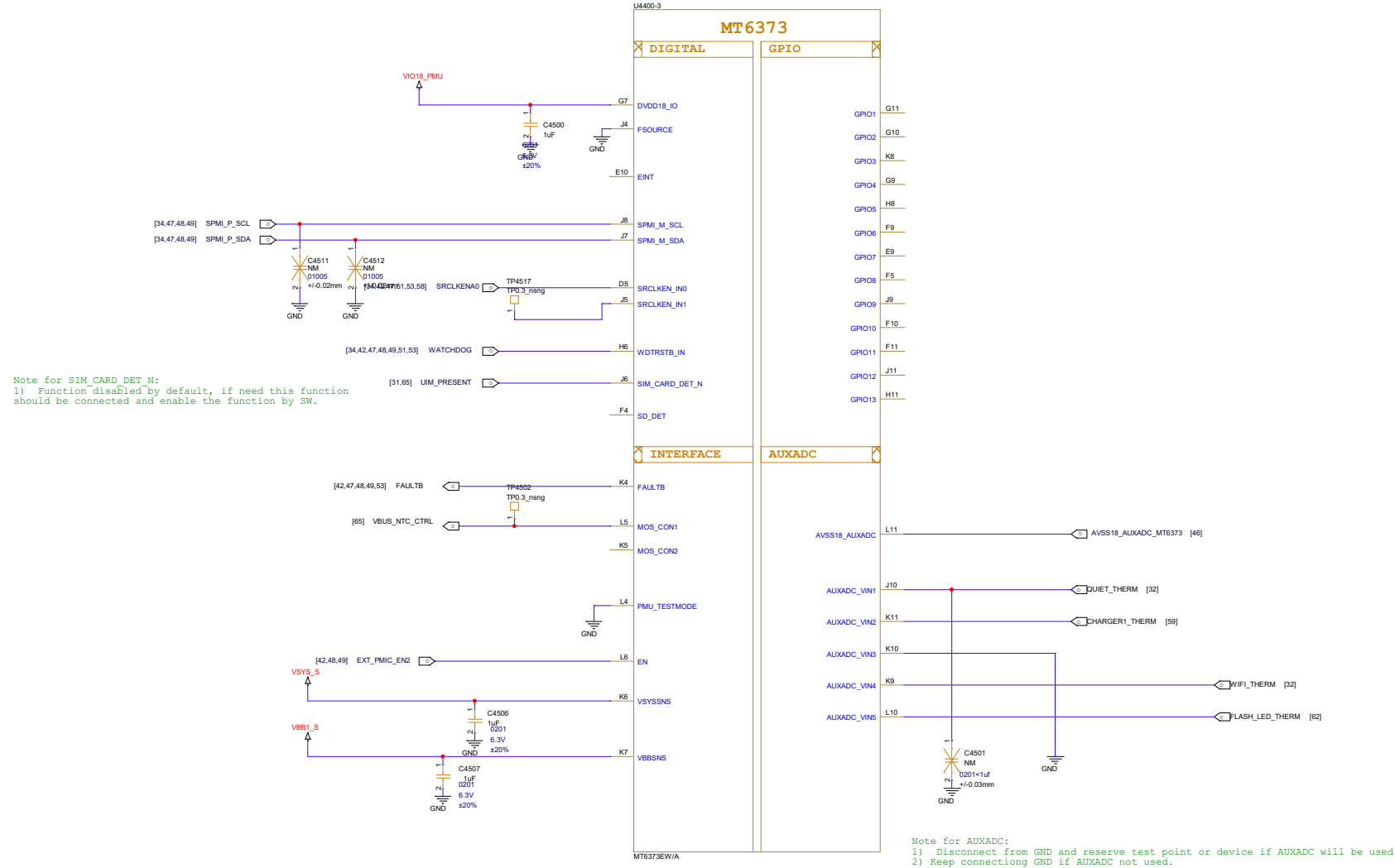
MT6195:all components in
input sig: 500-pad fixity.







Note for SIM_CARD_DET_N:
1) Function disabled by default, if need this function
should be connected and enable the function by SW.

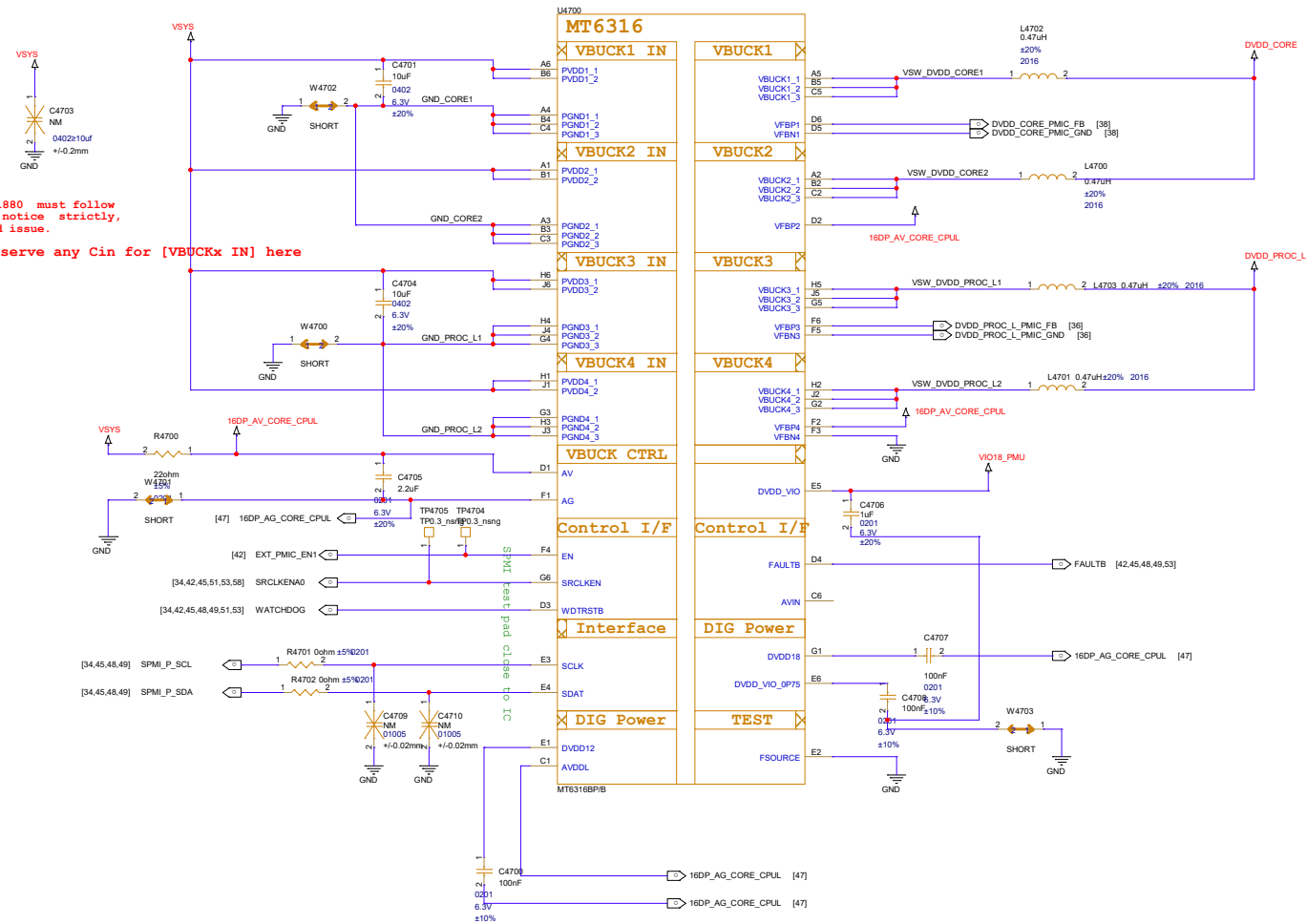


MT6316 4-Phase Buck (CORE_CPUL)

PMIC GND-ball connect to input cap. GND-pad firstly.

Warning: all Cin as C1877/C1880 must follow ref. design SCH and design notice strictly, reduction may cause damaged issue.

Warning: No need to reserve any Cin for [VBUCKx IN] here

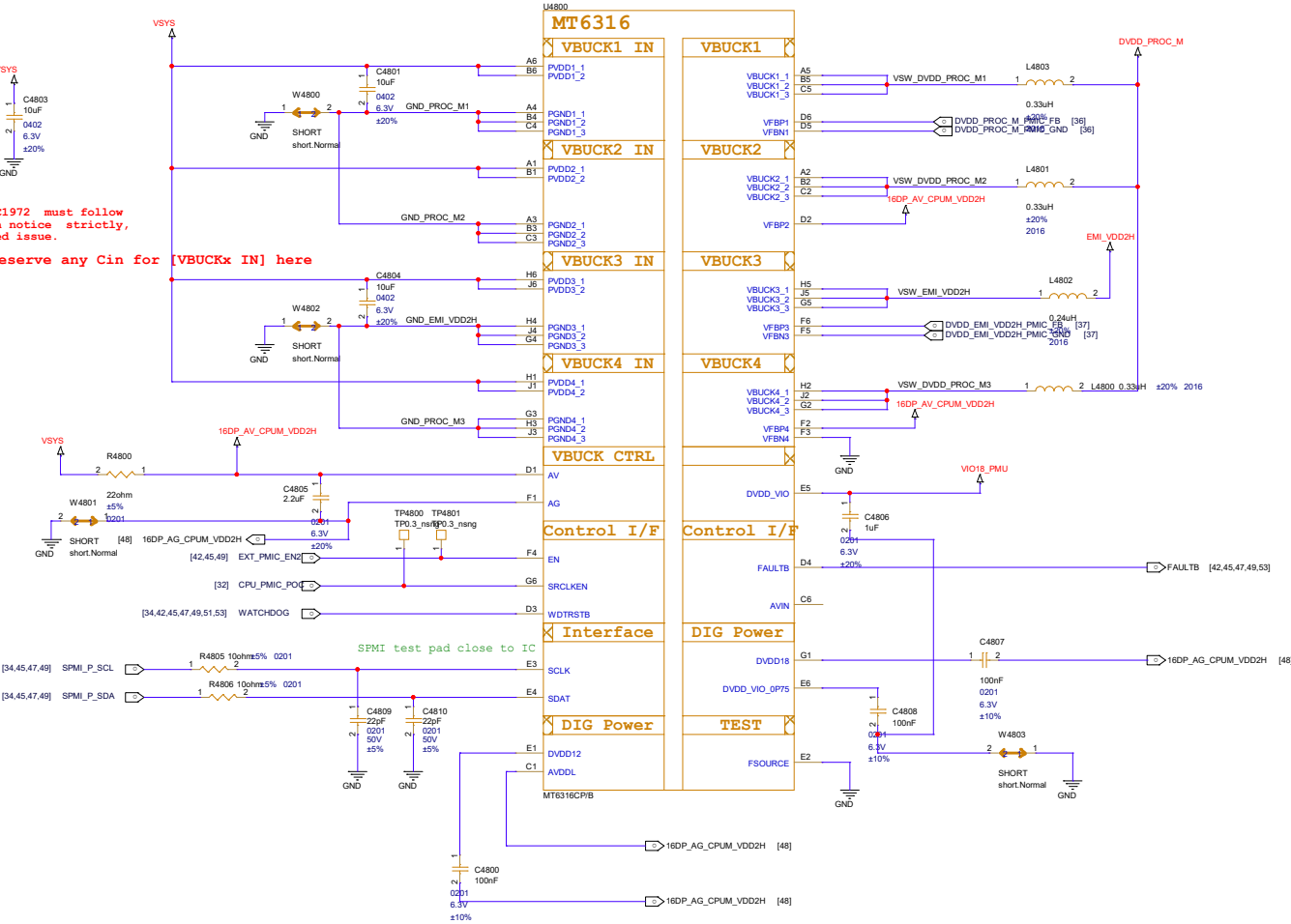


MT6316 4-Phase Buck (CPUM_VDD2H)

PMIC GND-ball connect to input cap. GND-pad firstly.

Warning: all Cin as C1969/C1972 must follow ref. design SCR and design notice strictly, reduction may cause damaged issue.

Warning: No need to reserve any Cin for [VBUCKx IN] here

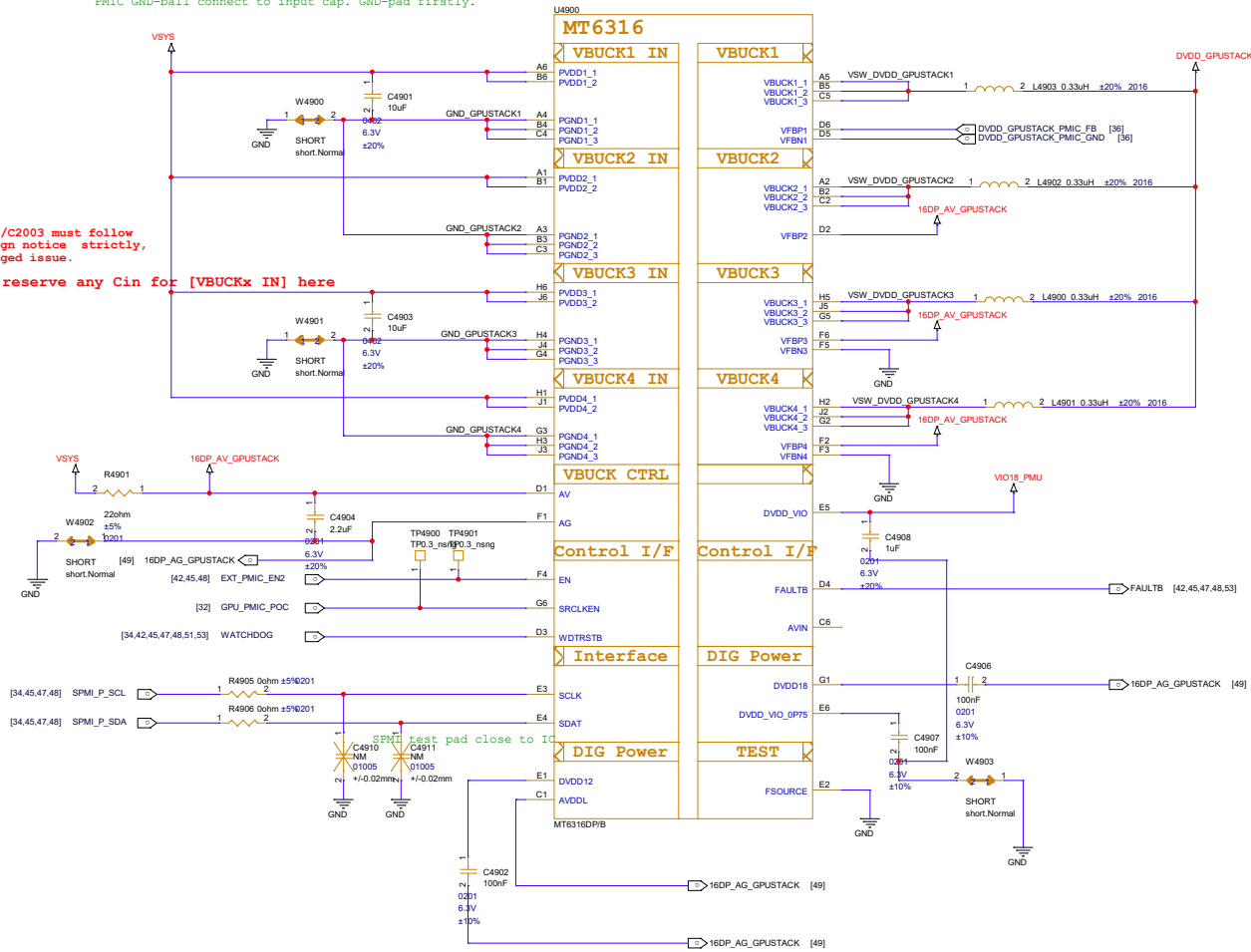


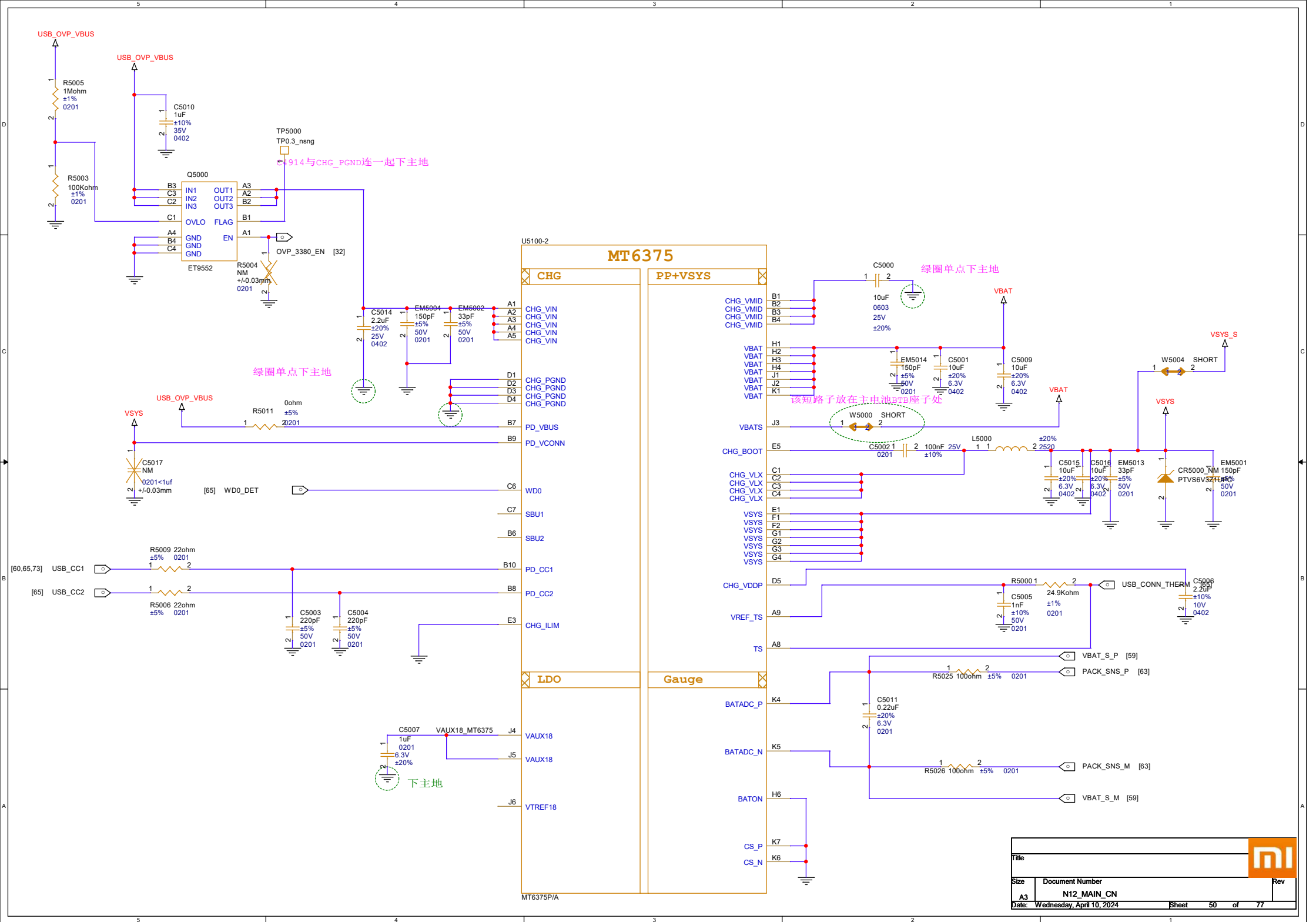
MT6316 4-Phase Buck (GPU STACK)

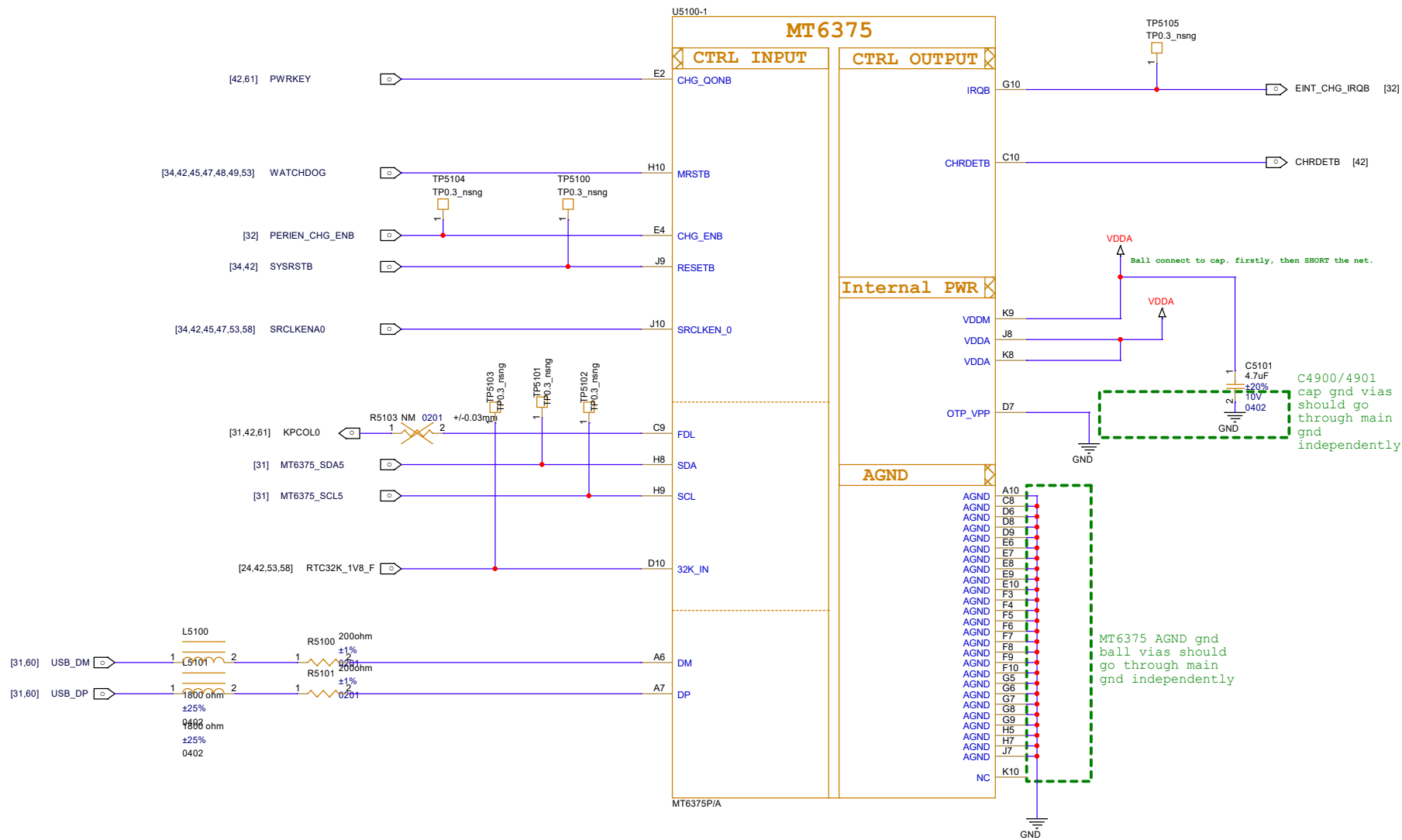
PMIC GND-ball connect to input cap. GND-pad firstly.

Warning: all Cin as C2001/C2003 must follow ref. design SCH and design notice strictly, reduction may cause damaged issue.

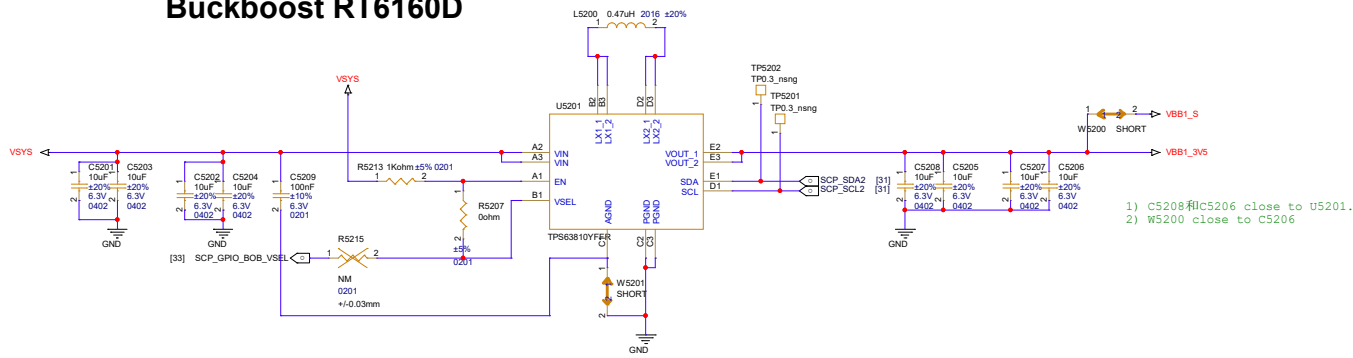
Warning: No need to reserve any Cin for [VBUCKx IN] here







Buckboost RT6160D



Schematic design notice of "85_MT6685 CLK AUXADC" page.

Note 85-1: MT6685 part # for 26M or 52M TMS selection

685 Part #	MT6685HP/B	MT6685LP/B
XTAL Freq	52M XTAL	26M XTAL

Note 85-2: DXCO CLK output assignment to CLK receiver

RF clocks		
Ball name	CLK Receiver	Freq.(MHz)
RFCK1A	FR2 RF / Misc.	XTAL (52/26)
RFCK1B	UFS4.0/3.1 combo	XTAL (52/26)
RFCK2A	FR1 (Sub-6G) RF	26M
RFCK2B	AP CK4 (FR2 ABB, GNSS)	26M

BB clocks		
Ball name	CLK Receiver	Freq.(MHz)
BBCK1	AP CK1 (System-boot)	26M
BBCK2	AP CK2 (WBG)	26M
BBCK3	Reserved for AP CK3 (UFS3.1)	26M
BBCK4	NFC	26M
BBCK5	Audio CODEC	26M

Note 85-3: Reserve 0R in series connection to form a R/C low pass filter for power noise reduction

Note 85-5: AUXADC application

Ball name	Application
AUXADC_VIN1	TMS
AUXADC_VIN2	AP temperature measurement in board level
AUXADC_VIN3	RF PA1 temperature measurement
AUXADC_VIN4	RF PA2 temperature measurement

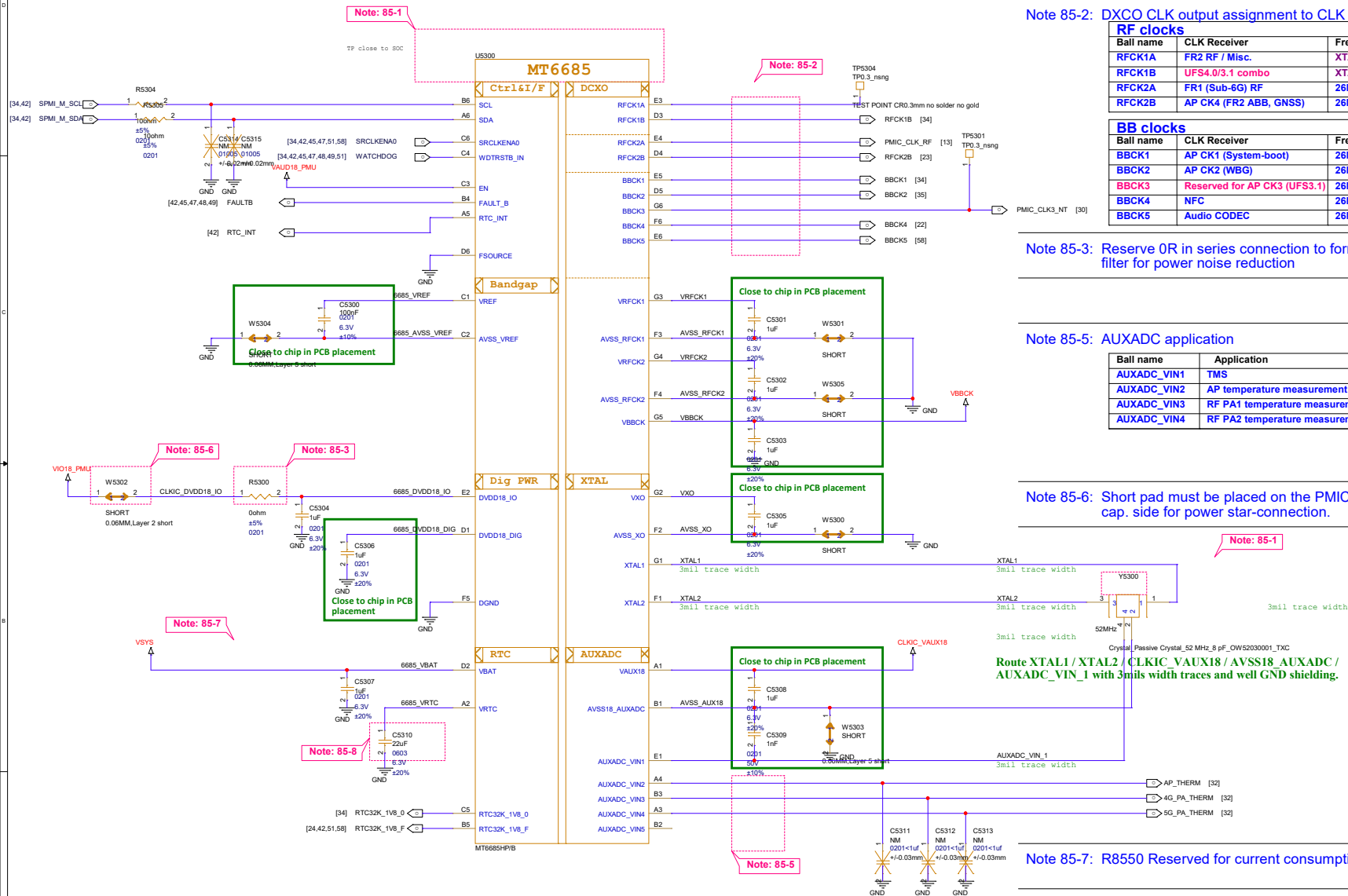
Note 85-6: Short pad must be placed on the PMIC VIO18 LDO's load cap. side for power star-connection.

Note: 85-1

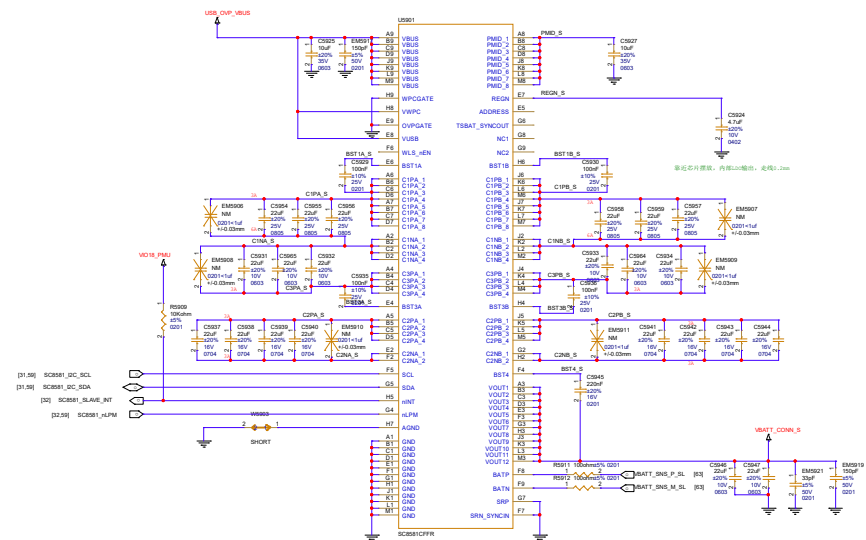
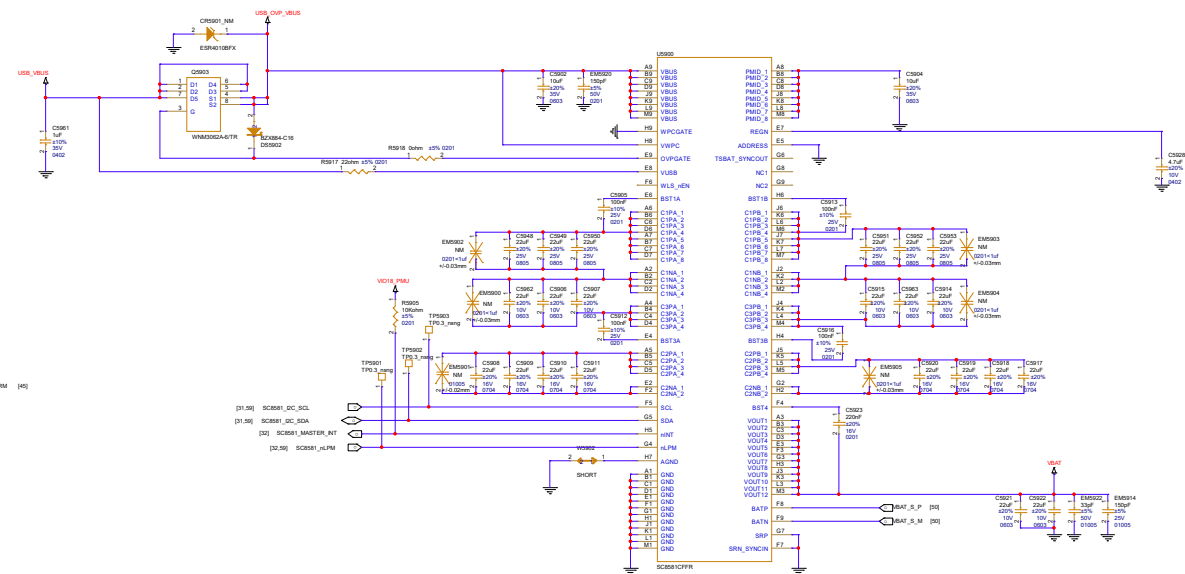
Route XTAL1 / XTAL2 / CLKIC_VAUX18 / AVSS18_AUXADC / AUXADC_VIN_1 with 3mils width traces and well GND shielding.

Note 85-7: R8550 Reserved for current consumption measurement.

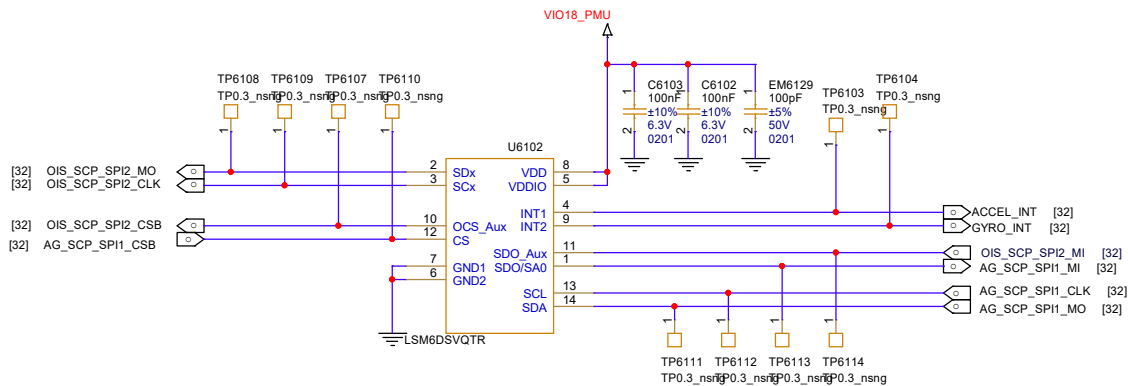
Note 85-8: Be aware the value selection for C8511 to support RTC data keeping and SPAR function.



Note: W



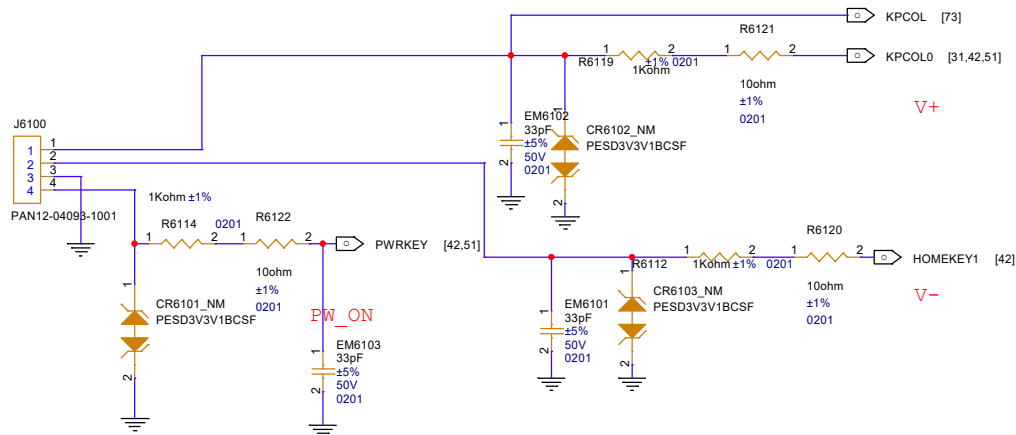
A+G



NOTE1:
如果OIS是四线SPI 贴R6108
如果OIS是三线SPI R6108 NM处理。

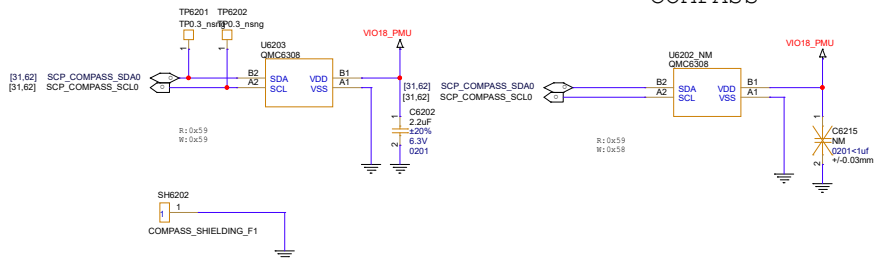
PRESSURE (气压传感器)

Sidekey



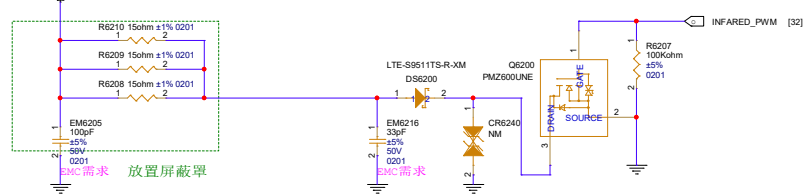
HALL (霍尔传感器)

COMPASS

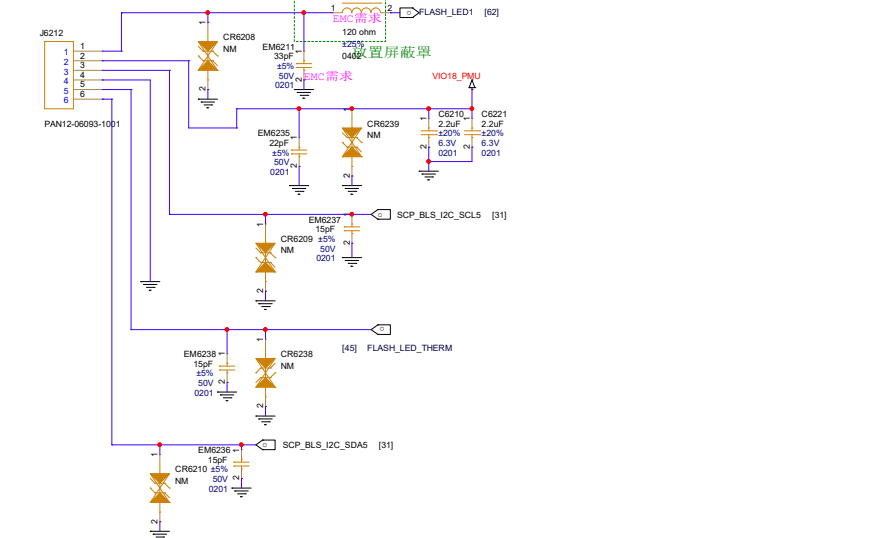


COMPASS

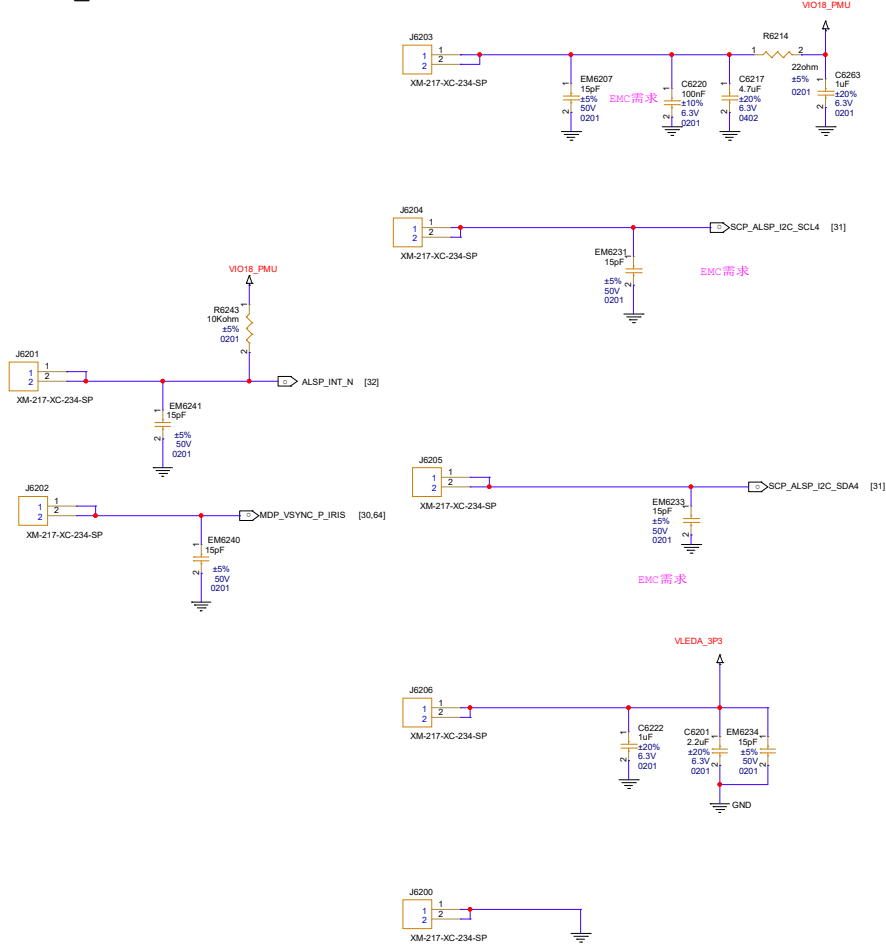
INFRARED



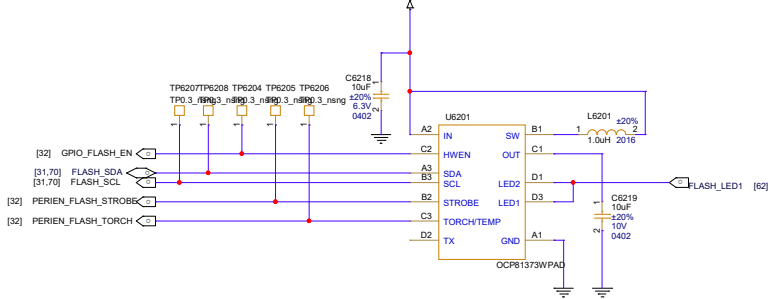
FLASH CON



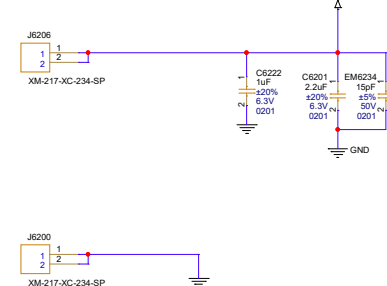
PL_SENSOR CON

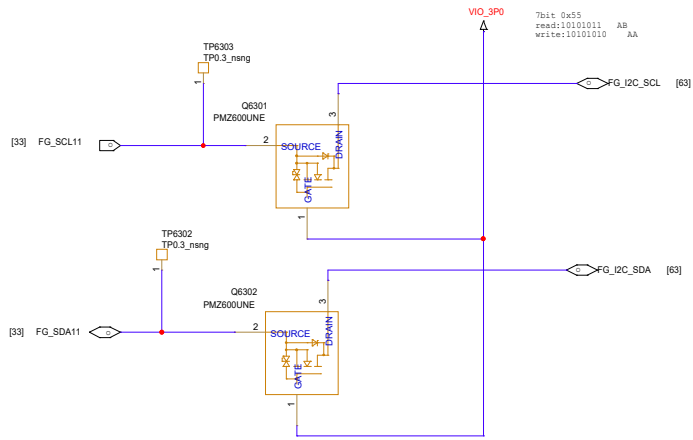


VSYS

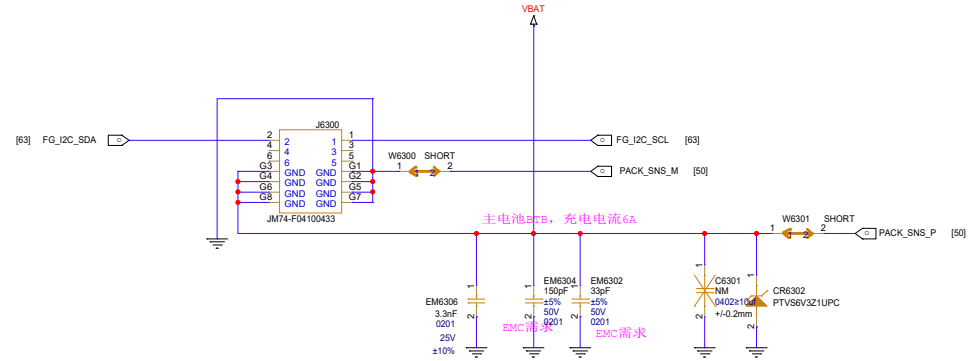


VLEDA_3P3

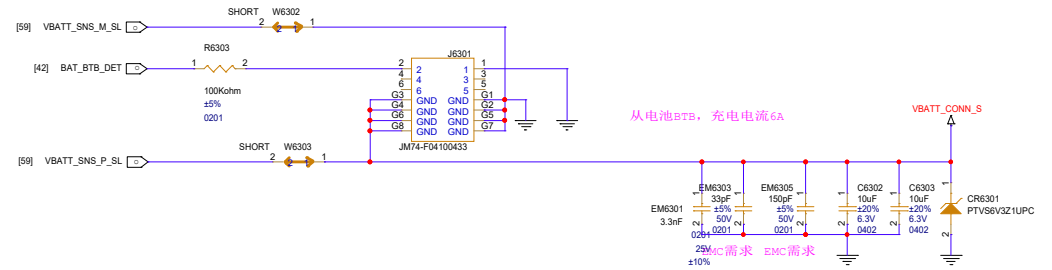


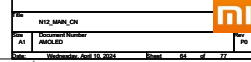
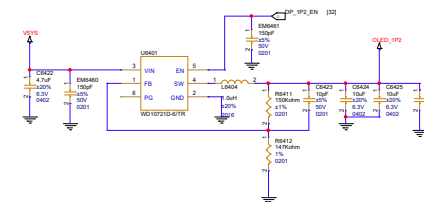
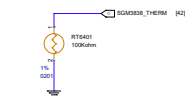


PRIMARY BAT CON

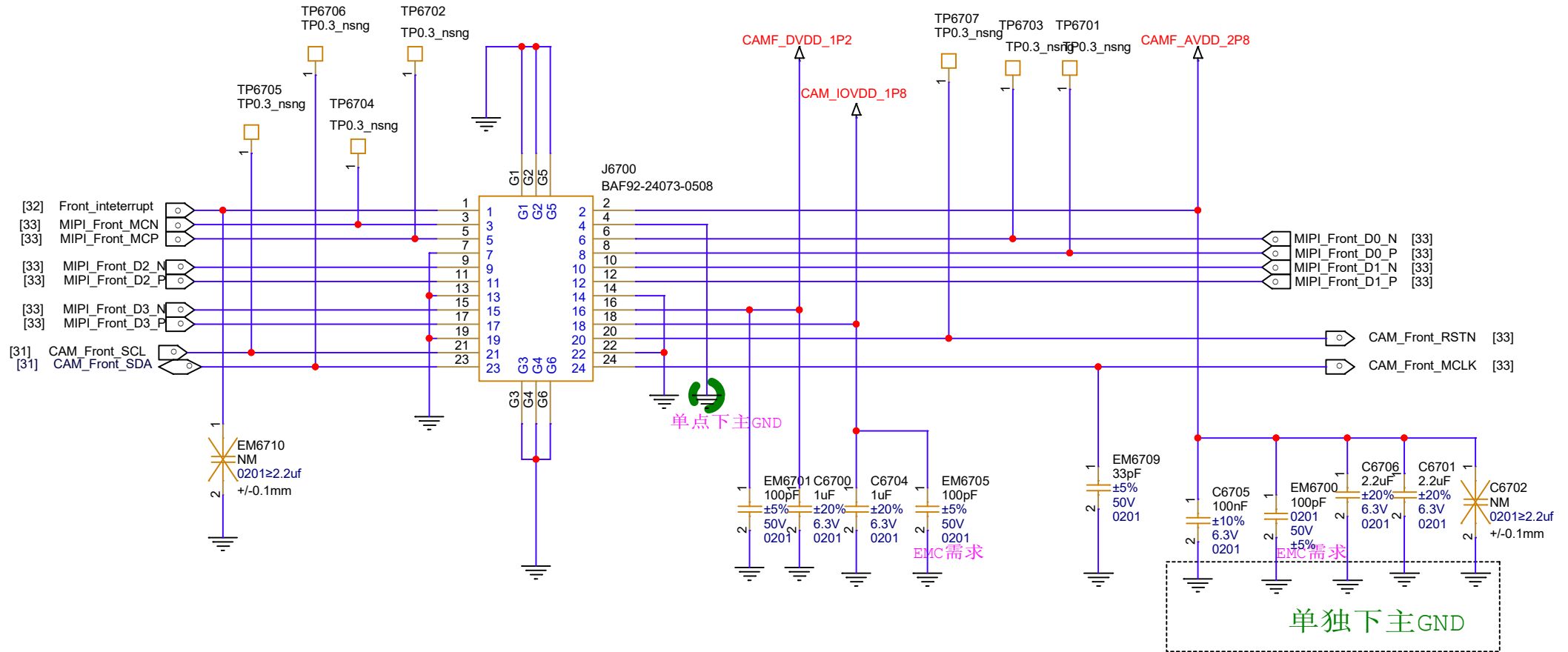


SECONDARY BAT CON

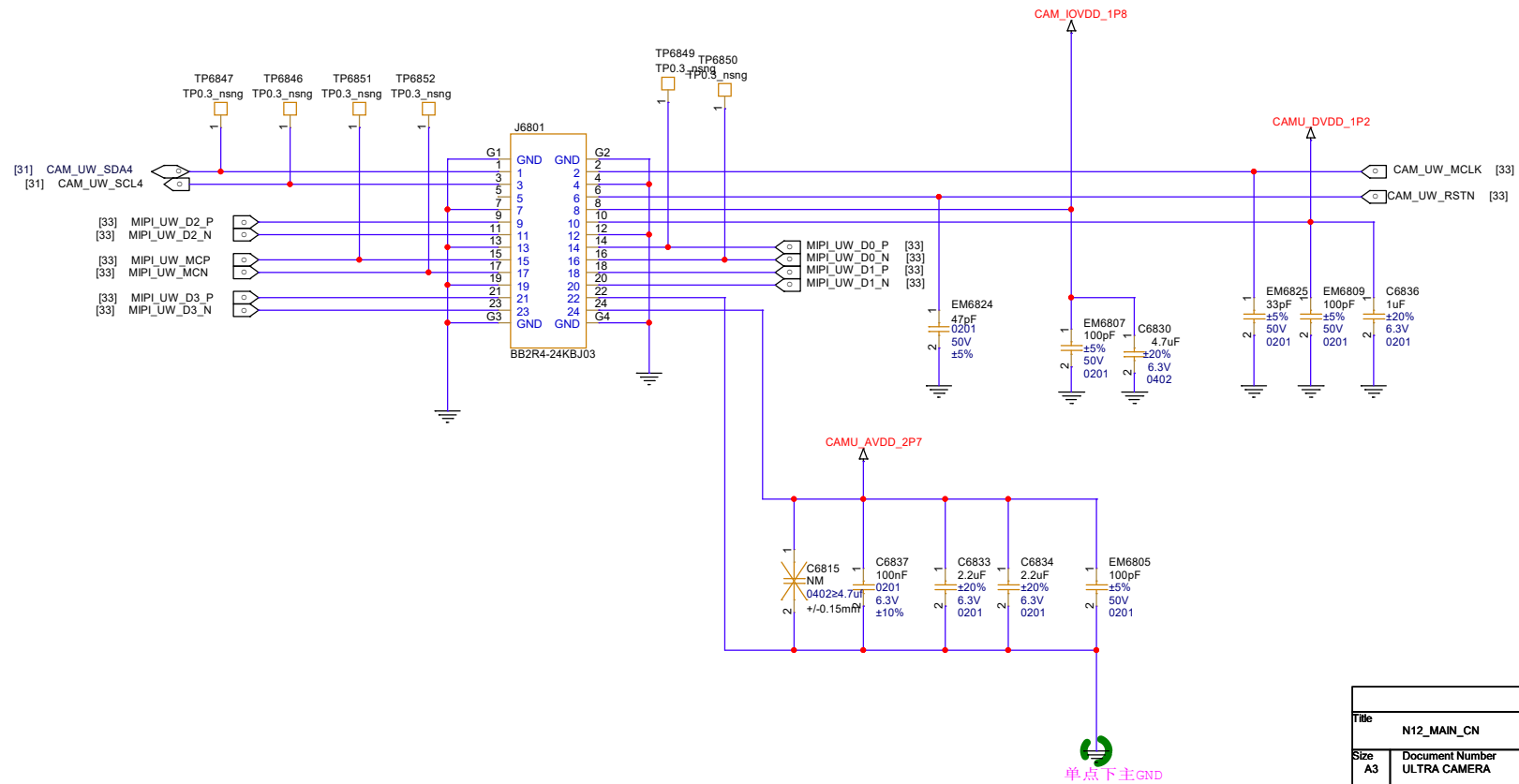




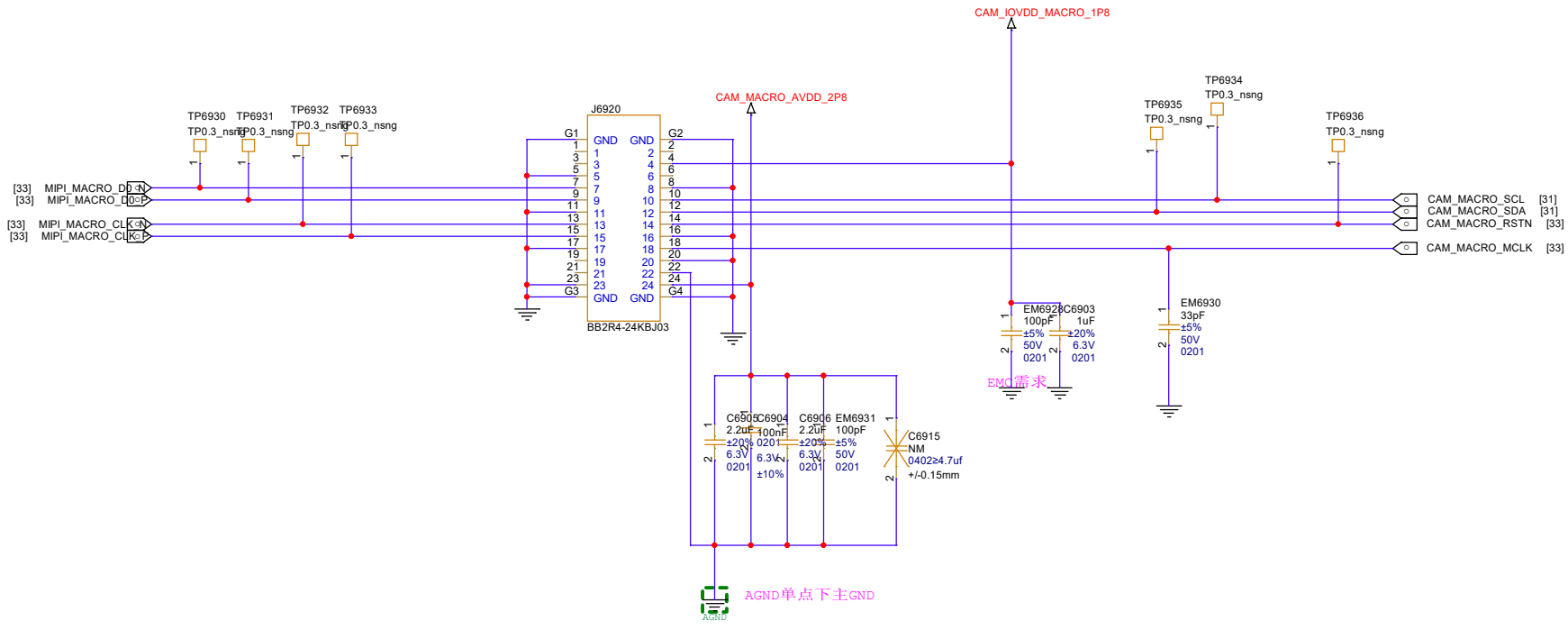
FRONT CAMERA 20M OV20B40



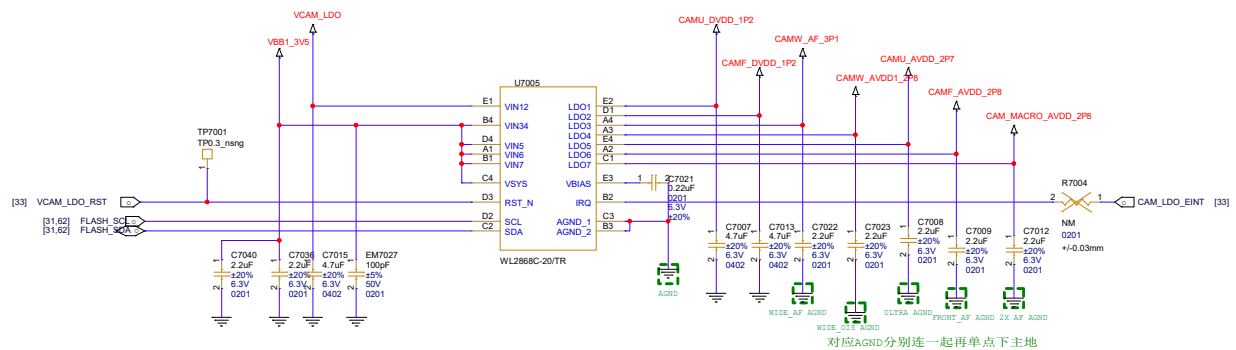
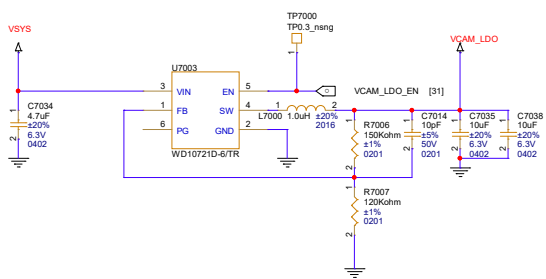
ULTRA 8M IMX355



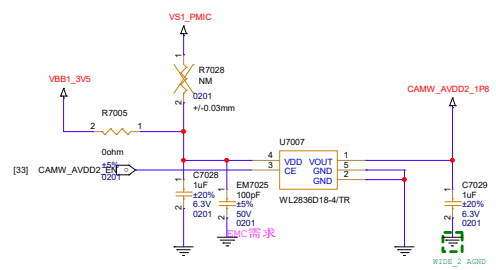
MACRO_2M GC02M1



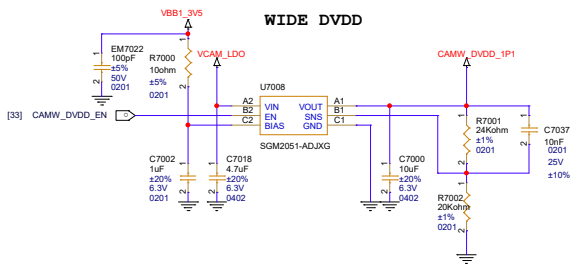
放置屏蔽罩



Close to WIDE and FRONT

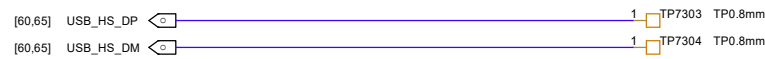
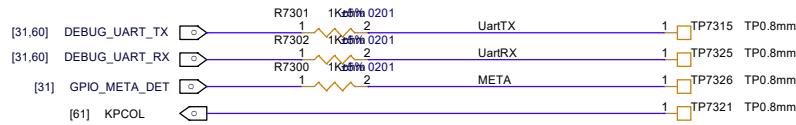
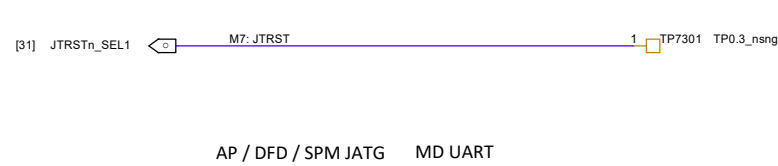


PIN	NET	VOL/V	CUR/mA
CAMW-VDDD	CAMW_DVDD_1P1	1.1	859/1206
CAMW-VDDIO	CAM_I0VDD_1P8	1.8	3/3.5
CAMW-VDDA1	CAMW_AVDD1_2P8	2.8	83/99
CAMW-VDDA2	CAMW_AVDD2_1P8	1.8	62/74
CAMW-VDDAF	CAMW_AF_2P8	2.8	120/120
CAMW-VDDOIS	CAMW_OIS_2P8	2.8	330/330
CAMU-VDDD	CAMU_DVDD_1P2	1.25	78/130
CAMU-VDDIO	CAM_I0VDD_1P8	1.8	0.4/3
CAMU-VDDA	CAMU_AVDD_2P8	2.8	35.7/45
CAM2X-VDDD	CAM2X_DVDD_1P05	1.05	200.54/260.7
CAM2X-VDDIO	CAM_I0VDD_1P8	1.8	2/4
CAM2X-VDDA	CAM2X_AVDD_2P8	2.8	56.42/67.7
CAM2X-VDDAF	CAM2X_AF_2P8	2.8	120/120
CAMF-VDDD	CAMF_DVDD_1P1	1.1	196.1/273.9
CAMF-VDDIO	CAM_I0VDD_1P8	1.8	1.8/2.9
CAMF-VDDA	CAMF_AVDD_2P9	2.9	50.3/72.9

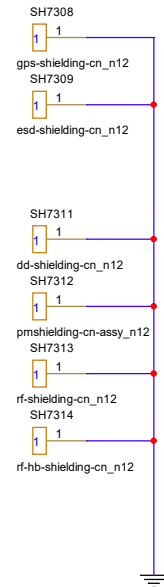
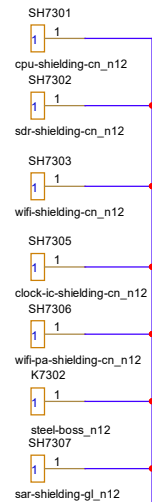
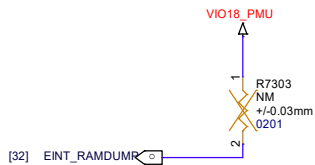


Schematic design notice of "90_DEBUG_IO" page.

Note 90-1: UART Default support : UART0,
JTAG Default support : AP JTAG/MD_IO_JTAG

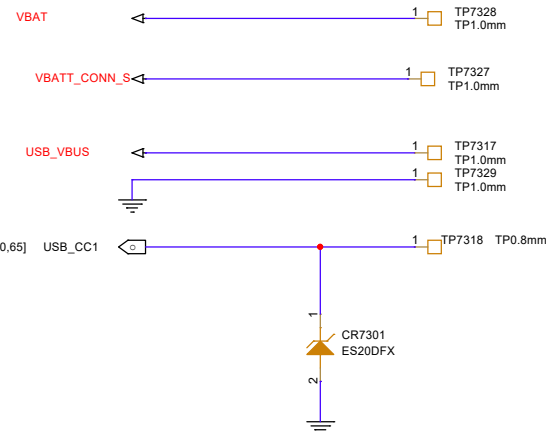


RAMDUMP debug key

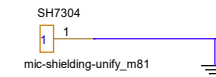


MD IO JTAG / ADSP_JTAG0 / BGF_DSP_L1

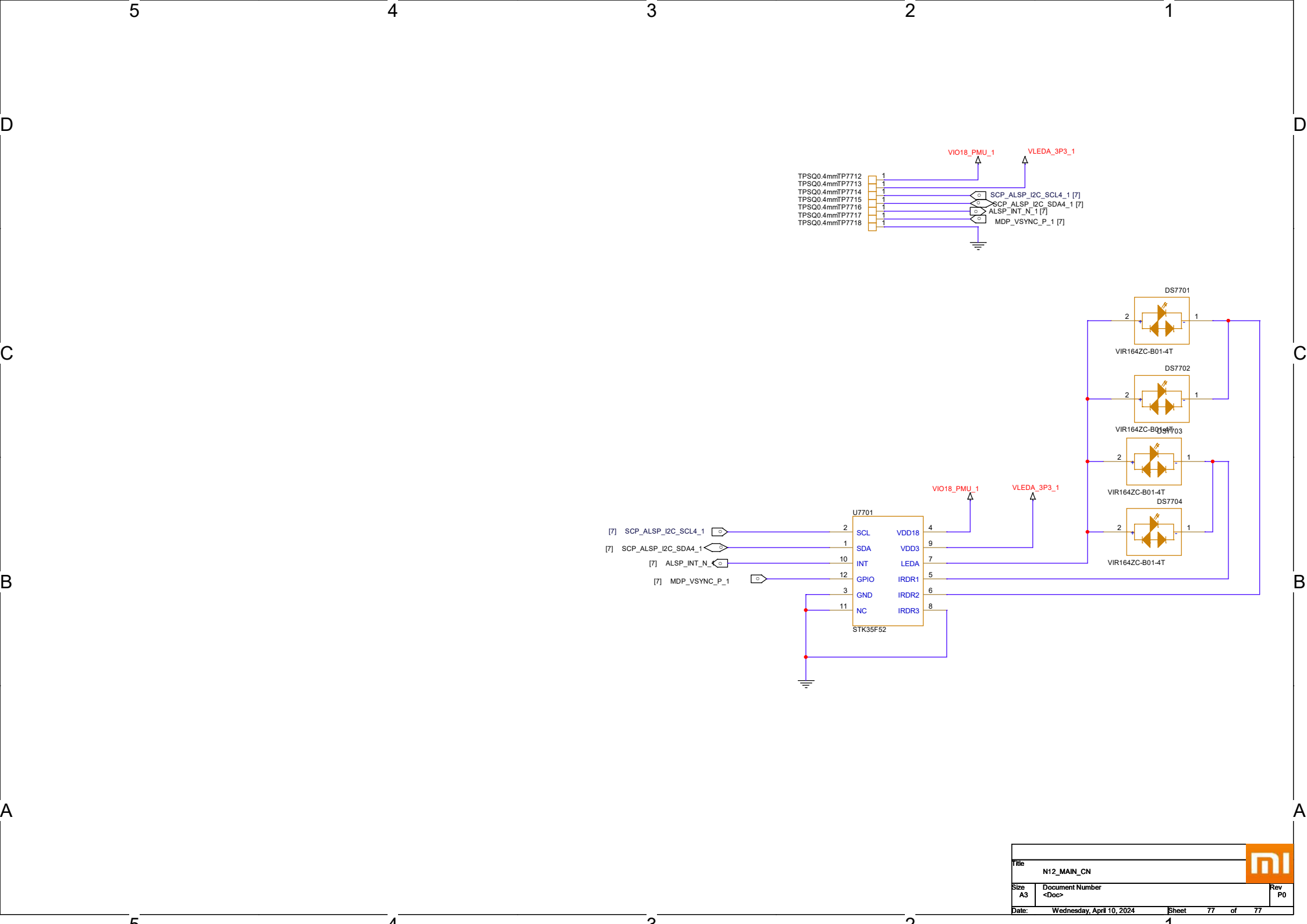
FORCE BOOT



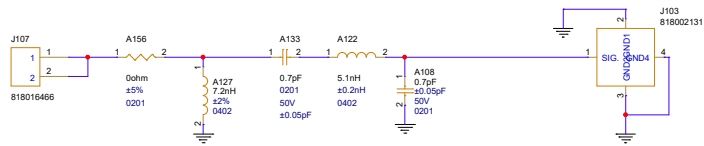
MIC 屏蔽罩



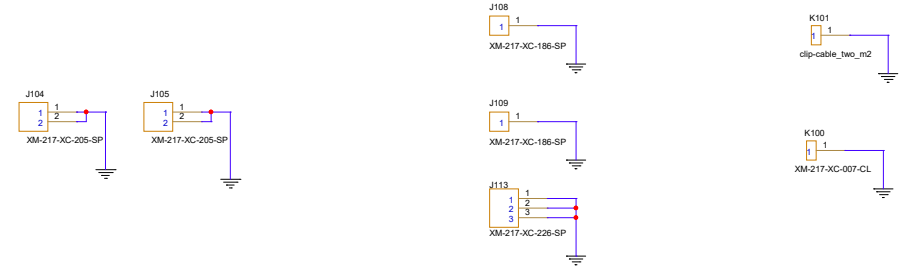
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Size	A3	Document Number	<Doc>	
Date:	Wednesday, April 10, 2024		Sheet 73 of 77	



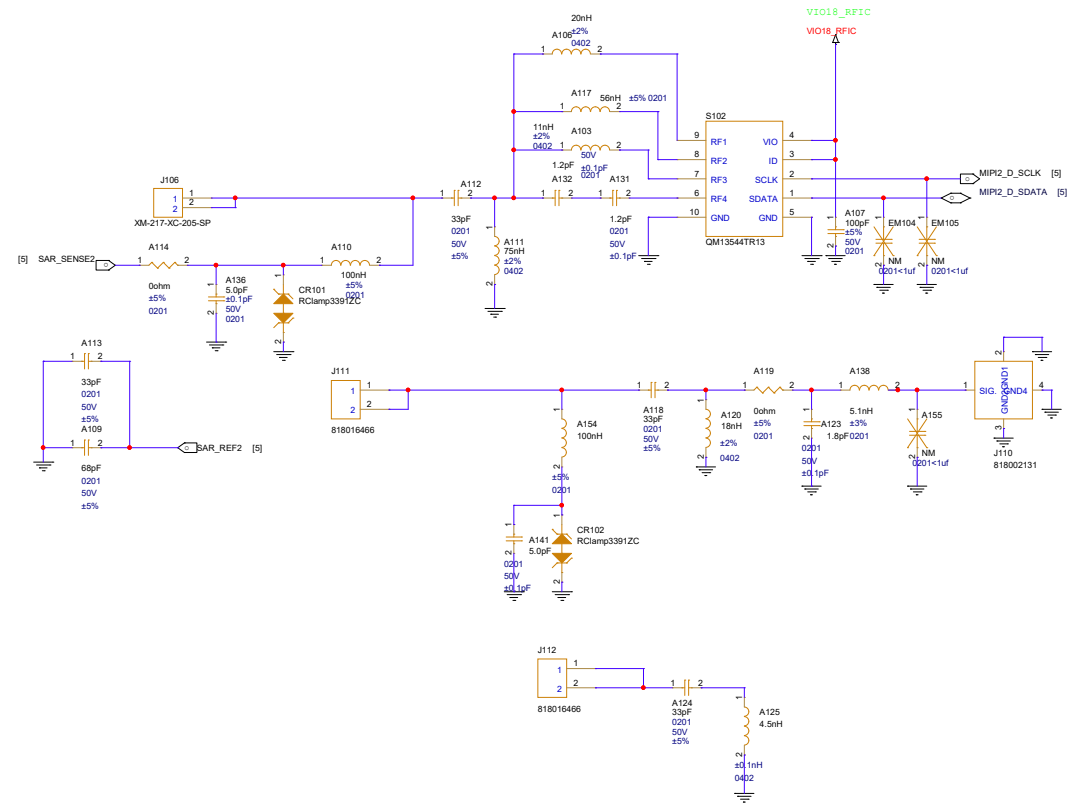
ANT5 MHB



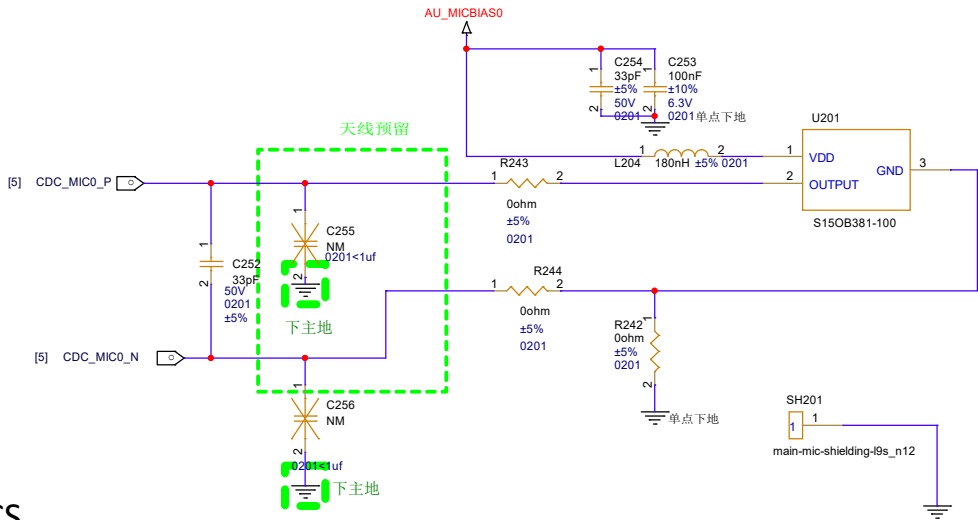
SUB_PCB_GND



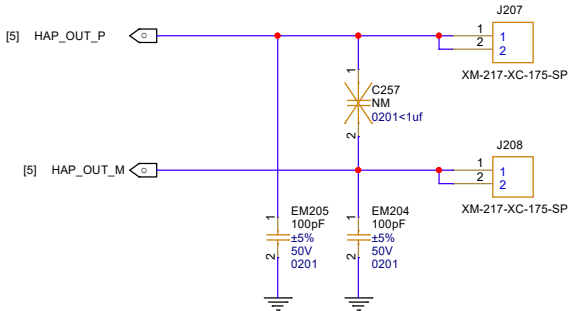
ANT0 LB_B32



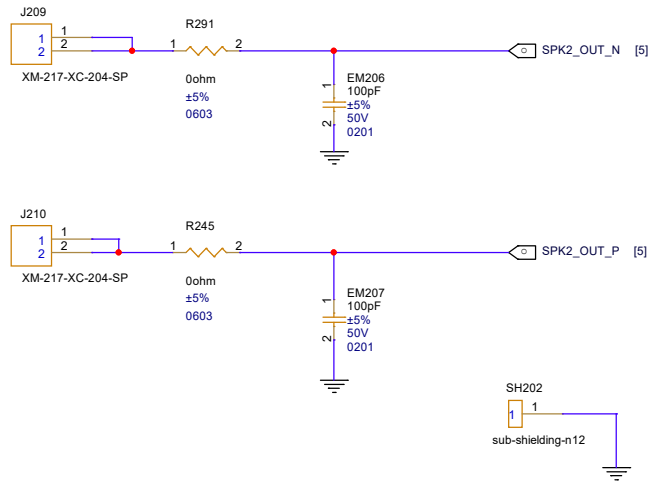
MIC

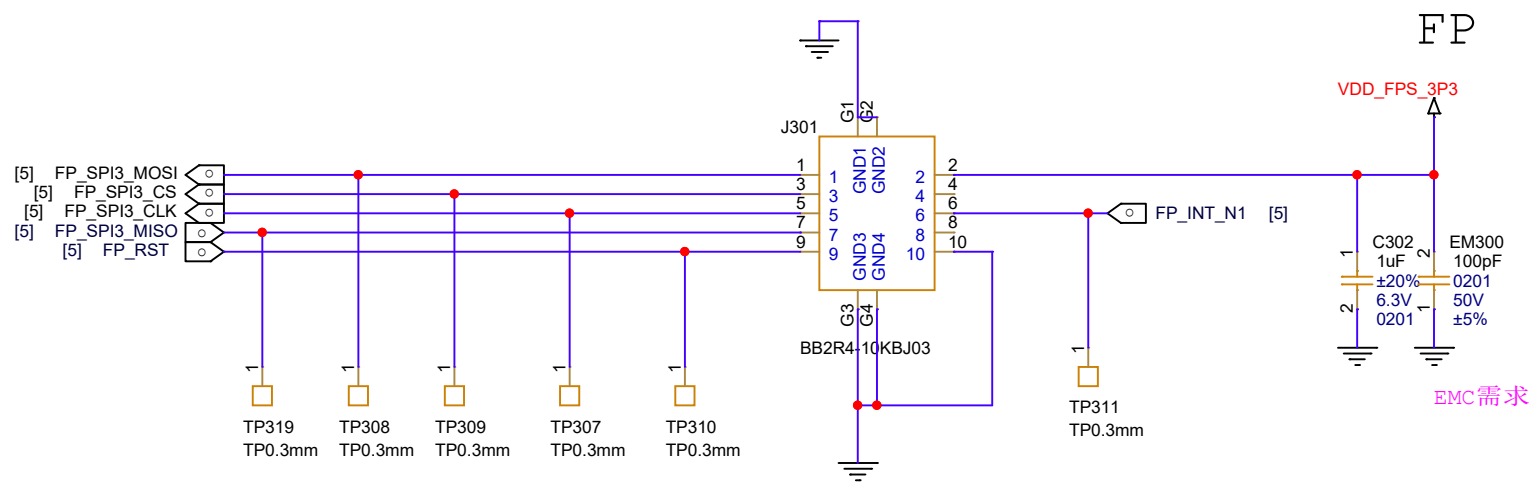


HAPTICS

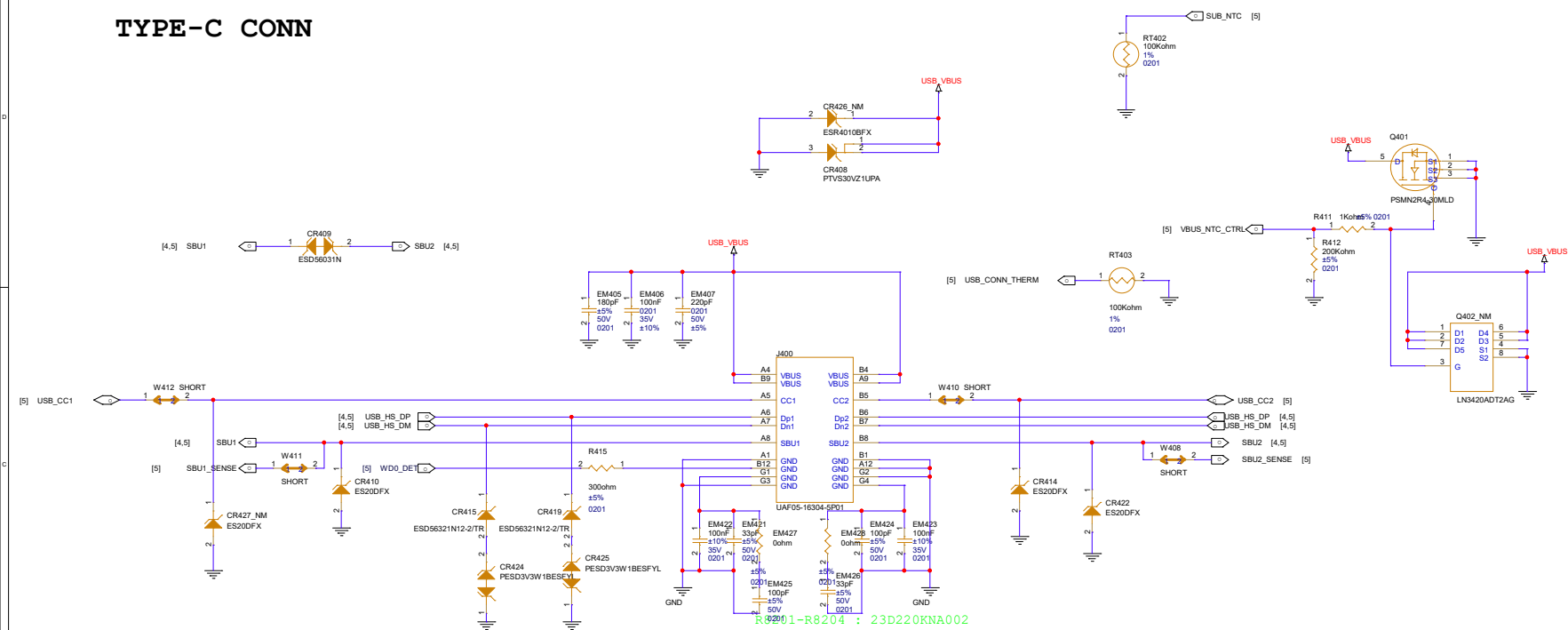


SPEAKER

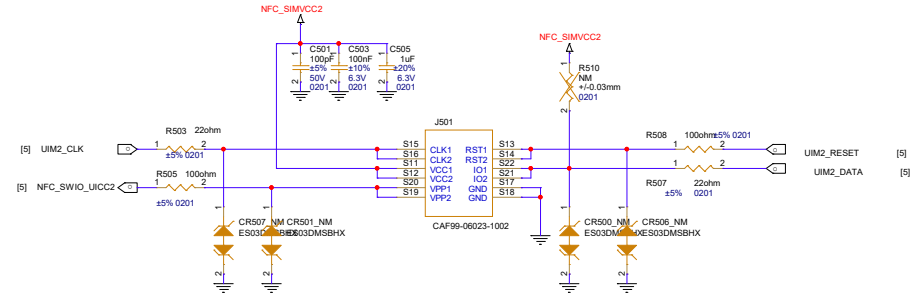
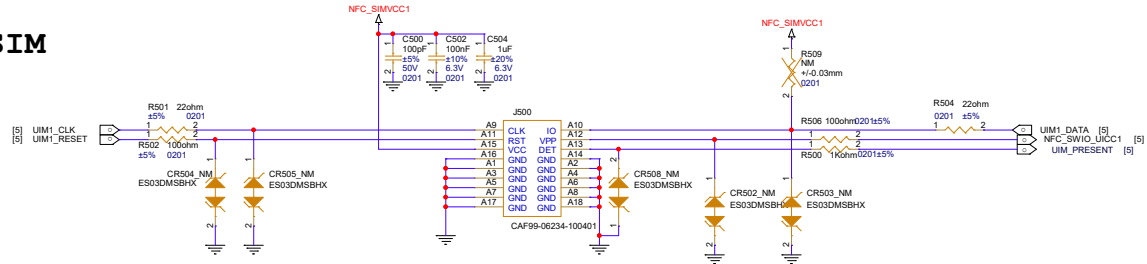




TYPE-C CONN



SIM



BTB CONNECTOR

